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La-induced microstructural regulation and residual stress mitigation in vacuum-brazed Al₂O₃ ceramic/AgCuTi/316 stainless steel joints



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Abstract

The influence of lanthanum (La) addition on microstructural evolution and residual stress mitigation in vacuum-brazed Al₂O₃ ceramic/AgCuTi/316 stainless steel (SS) joints was systematically investigated. The incorporation of La into the AgCuTi filler metal significantly improved its wettability and spreading behavior, resulting in a more homogeneous and refined interfacial microstructure. The joint containing 0.6wt% La demonstrated the best performance, with a 32.08% increase in shear strength, accompanied by a 73.66% reduction in compressive residual stress at the ceramic interface and a 58.76% decrease in tensile residual stress at the stainless steel interface. Detailed microstructural characterization revealed that La addition facilitated the

diffusion and uniform redistribution of Ti and Cu, effectively suppressing Ti segregation and minimizing the formation of the hard-brittle Ti-Cu intermetallic compounds, thereby promoting their refinement and dispersed distribution. This favorable microstructural evolution effectively alleviated the residual stress induced by the coefficient of thermal expansion mismatch between the joined materials. Therefore, the fracture path shifted from the Al₂O₃ ceramic to the brazing seam/ceramic interface, leading to enhanced shear strength of the joint.



Keywords

Vacuum brazing; Rare earth elements; Microstructure; Residual stress; Nanoindentation

1. Introduction

Al₂O₃ ceramics and 316 stainless steel (SS) are essential structural materials with broad applications in aerospace, chemical engineering, and related fields [[1], [2], [3], [4], [5], [6], [7]]. Despite their exceptional individual properties, their performance in extreme environments remains limited [8]: Al₂O₃ ceramics exhibit high brittleness and poor impact resistance [9], while 316 SS suffers from a significant loss of strength and oxidation resistance compared to ceramics under high temperature and severely corrosive conditions [10]. The development of ceramic-metal composite structures enables the combination of complementary advantages, such as the high-temperature stability and corrosion resistance of ceramics and the superior toughness of metals [[11], [12], [13]]. This approach is critical for achieving superior integrated performance in demanding environments.

However, the significant differences in the physical and chemical properties of ceramics and metals present three primary challenges in establishing reliable joints: (1) difficulty in achieving simultaneous wetting of both ceramic and metal surfaces; (2) formation of hard-brittle interfacial phases that deteriorate joint performance; and (3) joint failure induced by residual stress [[14], [15], [16]]. Currently, the primary joining techniques include brazing, self-propagating high-temperature synthesis (SHS) [17], and diffusion bonding [18]. Among these, active metal brazing (AMB) has become the preferred method for ceramic-to-

metal bonding due to its effectiveness in bonding dissimilar materials, operational simplicity, versatility, and wide applicability [19]. AgCuTi active brazing filler metals are widely used because of their excellent ductility and superior wettability on ceramic surfaces [20]. Nevertheless, despite the improved surface wetting afforded by active elements, challenges related to the formation of hard-brittle intermetallic compounds and the development of residual stress during the joining process persist and require further investigation [[21], [22], [23]].

Numerous studies have explored methods to improve joint performance. Qie et al. [24] incorporated TiC particles into Ag-Cu-Ti filler metal when brazing AlN to Cu. The addition of TiC effectively refined its microstructure and alleviated the thermal mismatch of the joints, ultimately increasing the shear strength by 81% to 120.9MPa. Li et al. [25] used a Cu-Ti/Cu/BNi-2 composite filler to join Al₂O₃ and Kovar alloy. The addition of Cu/BNi-2 effectively suppressed the formation of brittle Fe-Ti phases within the joint, and the Cu foil contributed to alleviating residual stresses. When the thickness of the Cu layer was 100μm, the joint shear strength reached 128±7MPa, representing a 50.5% improvement. In another study, Wang et al. [26] developed a surface honeycomb modification for the Cf/SiC composite via pre-oxidation, and then brazed it to 304 stainless steel using AgCuTi filler. This created an infiltration area with a nailing/pinning effect that dispersed residual stress, resulting in a maximum joint shear strength of 158MPa, several times higher than without modification. Li et al. [27] developed a novel surface texturing method to improve the reliability of laser weld-brazed aluminum/stainless steel joints, leading to a line load of 495.9N/mm—61% higher than that of untextured joints.

In recent years, research has increasingly focused on improving joint properties by optimizing filler metal composition and precisely controlling brazing parameters. Yao et al. [28] eliminated excessive Ti_xCu_y and Ti_xFe_y phases in TA1/304 SS joints by selecting AgCuNi filler metal. Their findings indicated that holding time influenced the formation and distribution of the Ti₃Cu₄ phase, while brazing temperature affected the formation of TiCu₄ and Ti_xFe_y phases, ultimately enabling high-quality bonding between TA1 and 304 SS. He et al. [29] incorporated the rare earth element Dy into Ti-15Cu-15Ni filler metal and refined the brazing process to join TC4 alloy, achieving a maximum shear strength of 894.8MPa. While particulate additives (e.g., ZrO₂, TiC) have been extensively studied for modifying brazing filler metals, they typically function as passive, inert reinforcements that primarily hinder crack propagation. Importantly, such additives do not engage actively in interfacial reactions and often impair wettability by increasing the molten filler viscosity. In contrast, rare earth elements have garnered considerable interest owing to their microalloying capabilities, such as grain refinement, wettability enhancement, and suppression of brittle phase formation [30,31]. Distinct from particulate additions, rare earth elements like La exhibit unique electronic structures (4f orbitals) and high oxygen affinity, enabling them to dynamically interact with reactive elements (e.g., Ti) at the atomic scale. This suggests a

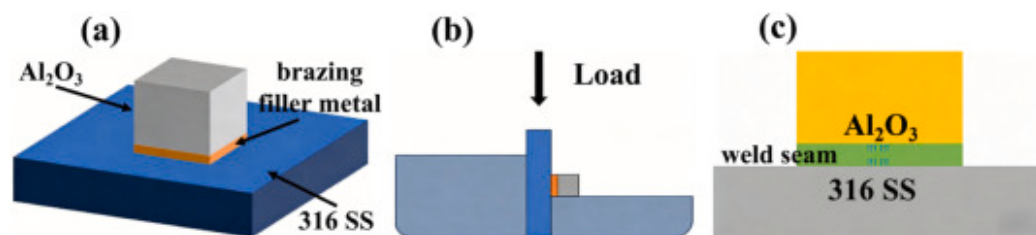
fundamentally different mechanism: rather than acting as inert fillers, La can actively modulate the thermodynamics and kinetics of interfacial reactions. However, existing research primarily addresses the microalloying effects of rare earth elements in bulk materials, with relatively limited research of their roles in brazing filler metals and brazed joints [32,33]. In particular, studies on the influence of rare earth elements on the microstructure and properties of ceramic/metal brazed joints remain scarce.

To address this research gap, this study investigated the vacuum brazing of Al₂O₃ ceramics and 316 SS using AgCuTi filler metal doped with varying amounts of La. The objectives were to identify the optimal La content and elucidate the underlying mechanisms of microstructural evolution and residual stress mitigation within the joint. The results demonstrated that La addition significantly modified the microstructure, mechanical properties, and residual stress profiles of the brazed joints. Furthermore, the intrinsic relationship between interfacial microstructure and fracture behavior was elucidated. Finite element simulations corroborated the proposed mechanisms, clarifying how the La-induced microstructural changes mitigated the development of residual stress.

2. Materials and methods

2.1. Materials

Al₂O₃ ceramics and 316 SS were selected as the base materials for brazing experiments. The filler metal used was an Ag-24Cu-4Ti (wt.%) paste, doped with x wt.% (x=0, 0.4, 0.6, 0.8) of La powder (99.99% purity), which was homogenized by mechanical stirring under an inert atmosphere. The Al₂O₃ ceramic specimens (99.9% purity) measured 5 mm×5 mm×5 mm and exhibited an average surface roughness of approximately 1 μm. The 316 SS substrates, polished progressively with SiC paper up to 2000 grit, had final dimensions of 16 mm×16 mm×4 mm. Prior to assembly, both substrates were ultrasonically cleaned in acetone to remove contaminants and ensure optimal brazing conditions. Approximately 20±0.5 mg of the filler metal was applied to the brazing surface of the Al₂O₃ ceramic, which was then assembled with the 316 SS substrate. A schematic diagram of the brazing assembly is shown in Fig. 1(a).



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Fig. 1. Schematic diagram of (a) specimen assembly, (b) shear test, and (c) nanoindentation indentation point locations.

2.2. Brazing processes

The assembled specimens were placed in a vacuum brazing furnace. To prevent oxidation of the filler metal during brazing, the process was conducted under a vacuum of 5×10^{-3} Pa. The heating procedure involved ramping at $10^\circ\text{C}/\text{min}$ to 250°C , holding for 30 min, then heating to 600°C and holding for 15 min. These holding steps were designed to remove moisture and organic residues from the filler metal, minimizing their adverse effects on brazing quality. The assembly was then held at 750°C for 15 min to ensure thermal uniformity and reduce temperature gradients. The temperature was subsequently raised to the brazing temperature of 900°C and held for 10 min to complete the joining process. To mitigate residual stress from rapid cooling, the furnace was cooled at a controlled rate of $5^\circ\text{C}/\text{min}$ to 650°C , after which the specimens were cooled to room temperature inside the furnace [34,35].

2.3. Characterization

The brazed joints were prepared for metallographic examination through standard procedures of mounting, grinding, and polishing, followed by ultrasonic cleaning in acetone. The microstructures of the brazed joints were characterized by scanning electron microscopy (SEM) with an energy-dispersive spectrometer, and X-ray diffraction (XRD). Shear strength tests were performed on the brazed joints using a universal testing machine at a constant rate of $0.5\text{ mm}/\text{min}$. To ensure statistical significance, three shear tests were performed for each brazing condition. A schematic of the shear test is shown in Fig. 1(b). After shear testing, the fracture morphology was analyzed by SEM, EDS and XRD to elucidate the fracture mechanism.

The micromechanical properties and residual stress of the brazed joints were tested at room temperature using an Agilent G200 nanoindentation system equipped with a Berkovich diamond indenter. A load of 5 mN was applied at the ceramic interface, while a load of 10 mN was applied at the stainless steel interface, with each load held for 10 s. The thermal drift was maintained below 0.05 nm/s, and a minimum spacing of 12 μ m was kept between adjacent indentations. To ensure data reliability, 20 to 30 tests were performed in each region, yielding Young's modulus, nano-hardness, and load-displacement curves. The specific indentation locations are shown in Fig. 1(c). Based on the load-displacement curves, residual stresses at both interfaces were calculated using the Suresh model [36]. An Ag-24Cu-4Ti brazing filler metal specimen, free of residual stress, served as the reference. Residual stresses under a fixed load can be calculated by the following equations:

$$\sigma = \frac{H}{\sin \alpha} \left(\frac{h_0^2}{h^2} - 1 \right) \text{ for compressive stress} \quad (1)$$

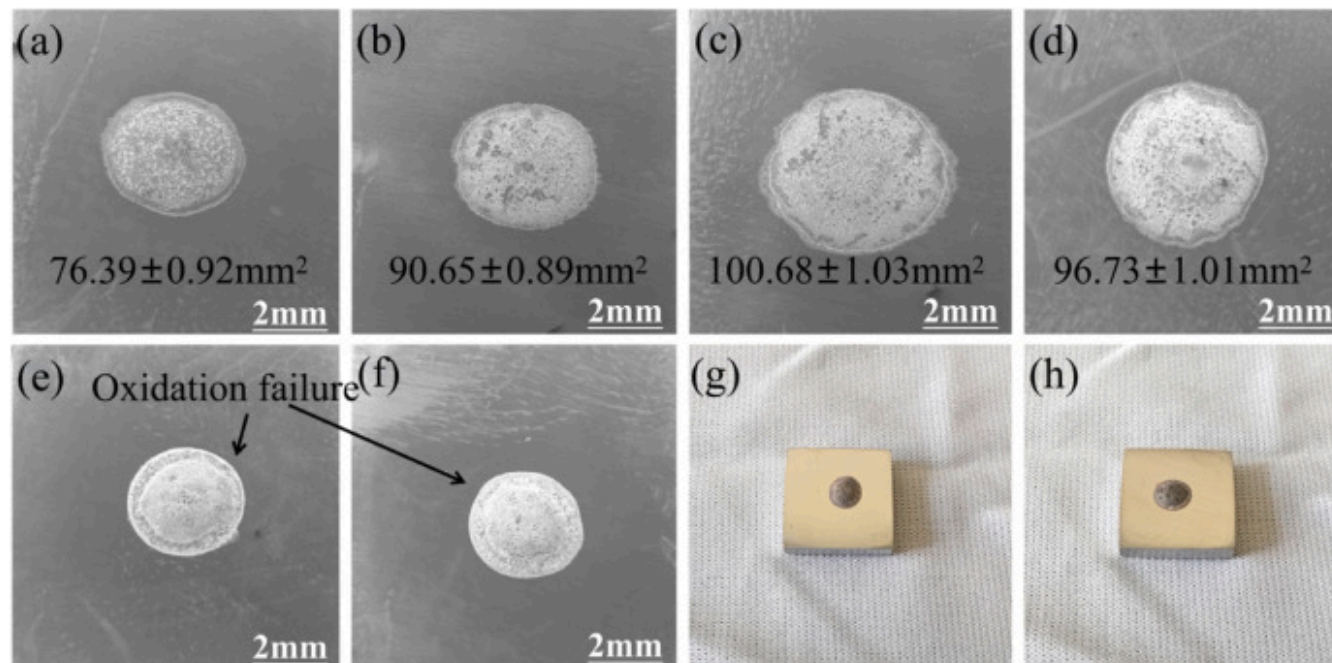
$$\sigma = H \left(1 - \frac{h_0^2}{h^2} \right) \text{ for tensile stress} \quad (2)$$

where H represents the nano-hardness of the material; h_0 and h denote the indentation depths under stress-free and measured conditions, respectively; and α is the half-angle of the indenter, with a value of 25.3°.

3. Results and discussion

3.1. Effect of La content on the wettability of AgCuTi-xLa filler metals

Good wettability is essential for achieving defect-free brazed joints. The adequate spreading of the molten filler metal can effectively fill the gaps and suppress the formation of pores, thereby enhancing the interfacial bonding strength [37]. Fig. 2 shows the spreading behavior of AgCuTi-xLa filler metals on 316 SS after holding at 900 °C for 10 min. It can be observed that the spreading area of the single filler metal was 76.39 mm². With the addition of 0.4 wt% and 0.6 wt% La, the spreading areas increased to 90.65 mm² and 100.68 mm², respectively. However, when the La content reached 0.8 wt%, slight oxidation occurred in the filler metal, reducing the spreading area to 96.73 mm², with a slight decline in wettability, but it remained superior to that of the single filler metal. Further increasing La content to 1 wt% and 2 wt% resulted in severe oxidation, causing the filler metal surface to turn grayish-black and deteriorating the wetting performance.



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Fig. 2. Spreading behavior of AgCuTi filler metals with different La contents: (a) 0wt%, (b) 0.4wt%, (c) 0.6wt%, (d) 0.8wt%, (e) 1wt%, (f) 2wt%; and macroscopic morphologies of the oxidized filler metals: (g) 1wt%, (h) 2wt%.

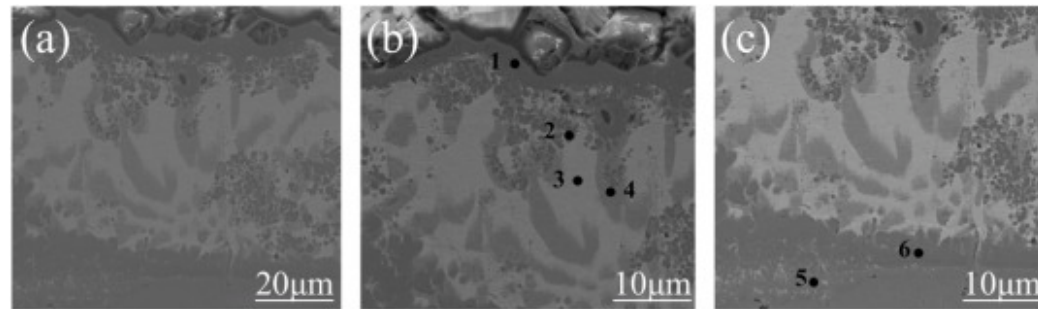
This non-monotonic variation reveals the dual mechanism of La in regulating wettability [38,39]. As a surface-active element with a vacant 4f orbital configuration (the strong delocalization of the 4f orbital electron cloud results in low interfacial adsorption energy of La) and high oxygen affinity, La preferentially segregates at the liquid-vapor interface through Gibbs adsorption [40]. According to the Gibbs adsorption isotherm: $\Delta\gamma_{LV} = -RT\Gamma \ln(1 + Kc)$ (where Γ is the surface excess concentration, R is the gas constant, T is the absolute temperature, K is the adsorption equilibrium constant, and c is the molar concentration of the solute), the interfacial enrichment of La reduces the liquid-vapor interfacial energy, directly enhancing the fluidity and spreading ability of the melt. Simultaneously, La acts at the solid-liquid interface, removing non-wetting oxide films such as Cr₂O₃ on the stainless steel surface through its strong oxygen affinity [41], and by dissolving into the Ti reaction layer (as shown by the co-distribution trend of La and Ti in Fig. 6, it optimizes the interfacial electron cloud distribution, thereby reducing

the solid-liquid interfacial energy. According to Young's equation: $\cos \theta = \frac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}}$ (where θ is the contact angle, γ_{sv} is the solid-vapor interfacial tension, γ_{sl} is the solid-liquid interfacial tension, and γ_{lv} is the liquid-vapor interfacial tension), the synergistic reduction of γ_{lv} and γ_{sl} significantly decreases the contact angle θ . Combined with the spreading area formula $A = k(1 - \cos \theta)$ (where k is a proportionality constant), a quantitative increase in spreading area is ultimately achieved.

It should be noted that the Ag-24Cu-4Ti single filler metal inherently exhibits good wettability due to its active Ti element and near-eutectic composition. However, the addition of La further increased the spreading area by 31.80% through the aforementioned dual interfacial regulation. Nevertheless, when the La content is ≥ 0.8 wt%, the adsorption concentration of La at the liquid-vapor interface exceeds the critical value, leading to a nonlinear rebound in interfacial energy. Concurrently, excess La forms a high-viscosity enrichment zone on the melt surface, which, through the dual effects of “physical barrier” and “kinetic inhibition”, hinders atomic diffusion and melt flow, ultimately deteriorating the wettability. This verifies the precise regulation effect of La on interfacial energy at critical concentrations.

3.2. Typical interfacial microstructure of the Al₂O₃/316 SS brazed joint with AgCuTi single filler metal

Fig. 3 shows the typical microstructure of the Al₂O₃/316 SS joint brazed with AgCuTi single filler metal at 900 °C for 10 min. As shown in Fig. 3(a), the brazing seam was completely filled with the filler metal, and no defects such as voids or microcracks were observed. The joint can be divided into three distinct regions: (I) the dark-gray reaction layer adjacent to Al₂O₃ ceramic; (II) the brazing seam composed of white and gray phases, with dark-gray particles distributed within; and (III) the diffusion region on the 316 SS side, consisting of light-gray and dark-gray granular constituents. The corresponding EDS composition analysis results for the different reaction phases in the joint are listed in Table 1. A continuous and uniform interfacial reaction layer with a thickness of $2.31 \pm 0.2 \mu\text{m}$ (point 1) was observed on the Al₂O₃ ceramic side in Fig. 3(b). This reaction layer was mainly composed of Ti and Cu. According to previous studies, this layer is most likely the Ti₃Cu₃O compound [42], which is also denoted as Ti₃(Cu,Al)₃O in some references. The continuous and uniform Ti₃Cu₃O interfacial reaction layer played a critical role in achieving effective bonding between the Al₂O₃ ceramic and the filler metal. An enrichment layer of the Ti₂Cu phase (point 2) was observed in the brazing seam near this reaction layer. The central region of the brazing seam was mainly composed of a white matrix phase of Ag (ss.) (point 3) and a gray matrix phase of Cu(ss.) (point 4). The diffusion region on the 316 SS side exhibited a mixed distribution of TiCu and TiFe phases (point 6), while the region adjacent to the stainless steel was predominantly composed of the TiFe₂ phase (point 5).



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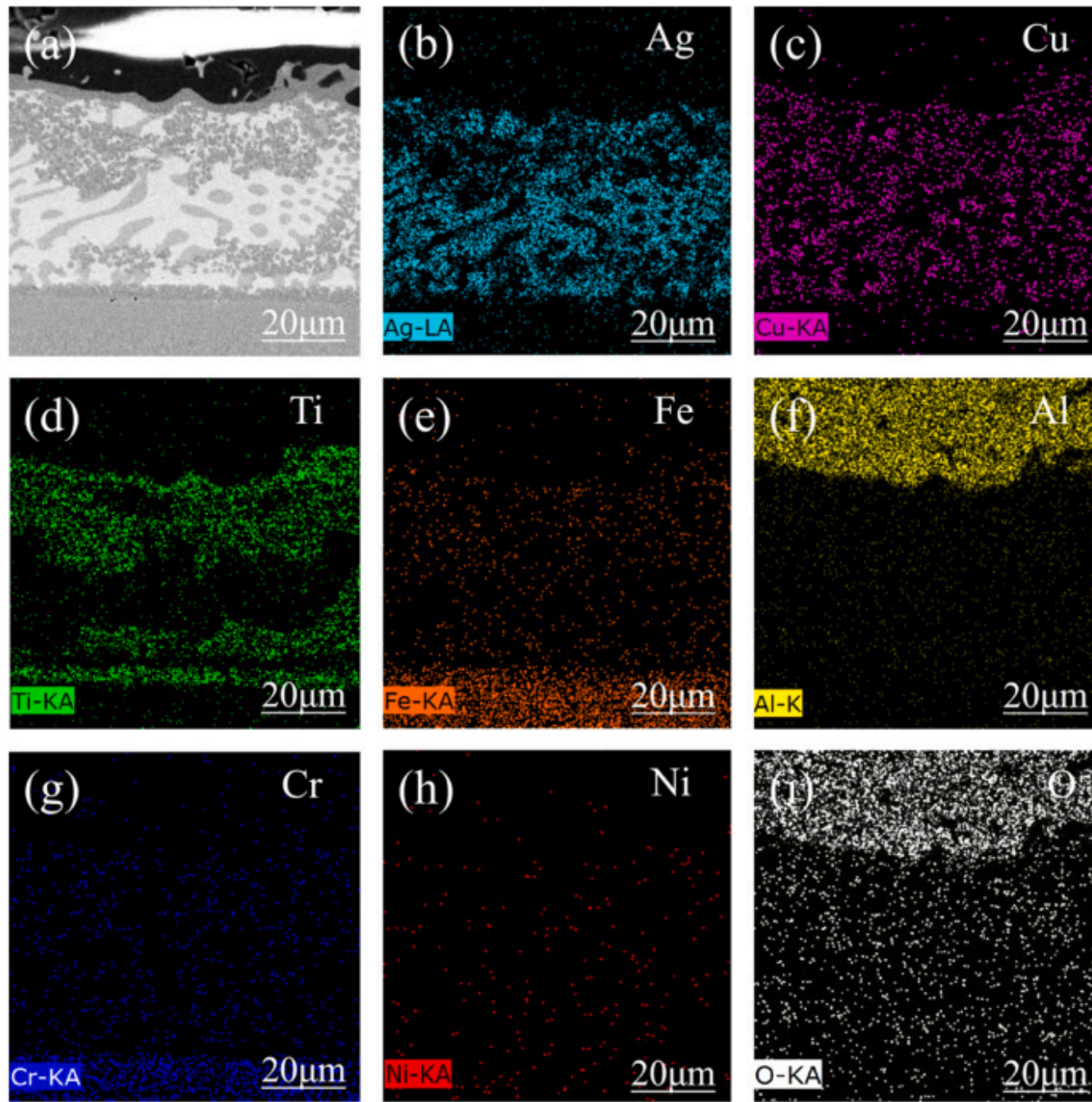
Fig. 3. Microstructures of joints brazed at 900°C for 10 min: (a) Al₂O₃/AgCuTi/316 SS joint; (b) Al₂O₃/AgCuTi interface; (c) AgCuTi/316 SS interface.

Table 1. EDS composition (at.%) results of the marked locations in Fig. 3.

Point	Ti	Cu	Fe	Ag	Al	Ni	Cr	La	Possible phases
1	44.73	43.69	-	-	6.33	-	-	-	Ti ₃ Cu ₃ O
2	63.78	30.94	-	2.33	2.46	-	-	-	Ti ₂ Cu
3	-	8.92	-	87.65	-	-	-	-	Ag (ss.)
4	-	96.75	-	3.25	-	-	-	-	Cu(ss.)
5	17.36	-	58.21	4.51	-	7.61	12.31	-	TiFe ₂
6	45.37	6.34	34.58	2.67	-	8.66	2.38	-	TiFe, TiCu

Fig. 4 presents the elemental mapping of the Al₂O₃/AgCuTi/316 SS joint, which verifies the above results. Ag (ss.) and Cu(ss.) constituted the matrix phases of the joint. A large amount of Ti was enriched at the ceramic surface, forming the interfacial reaction layer Ti₃Cu₃O, and an extensive Ti₂Cu phase was formed in the brazing seam near the Al₂O₃ side. The Fe, Cr, and Ni

elements from 316 SS substrate were uniformly distributed across the brazing seam, with no continuous band-like brittle intermetallic compounds (IMCs) forming within it [43]. This phenomenon was attributed to the solution treatment performed on the stainless steel substrate prior to brazing. According to Qian et al. [44], solution treatment eliminates lattice defects and reduces grain boundary area, thereby suppressing the diffusion of substrate elements into the brazing region and decreasing the content of brittle IMCs at the stainless steel interface.



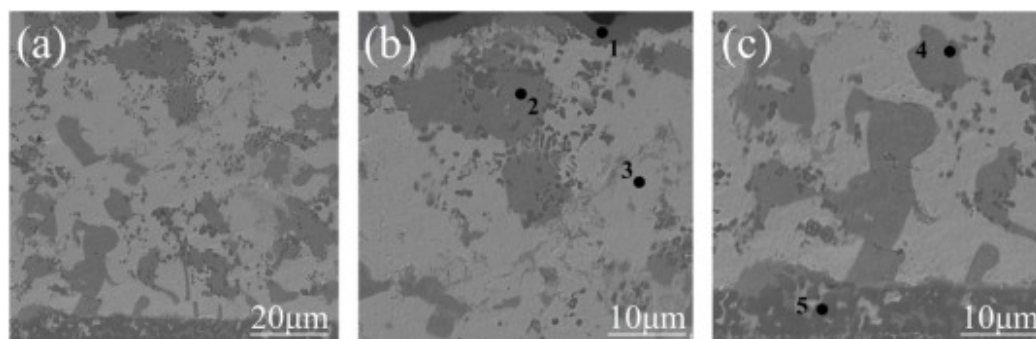
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Fig. 4. Elemental mappings of the Al₂O₃/AgCuTi/316 SS joint brazed at 900°C for 10min.

3.3. Typical interfacial microstructure of the Al₂O₃/316 SS brazed joint with AgCuTi-0.6La composite filler metal

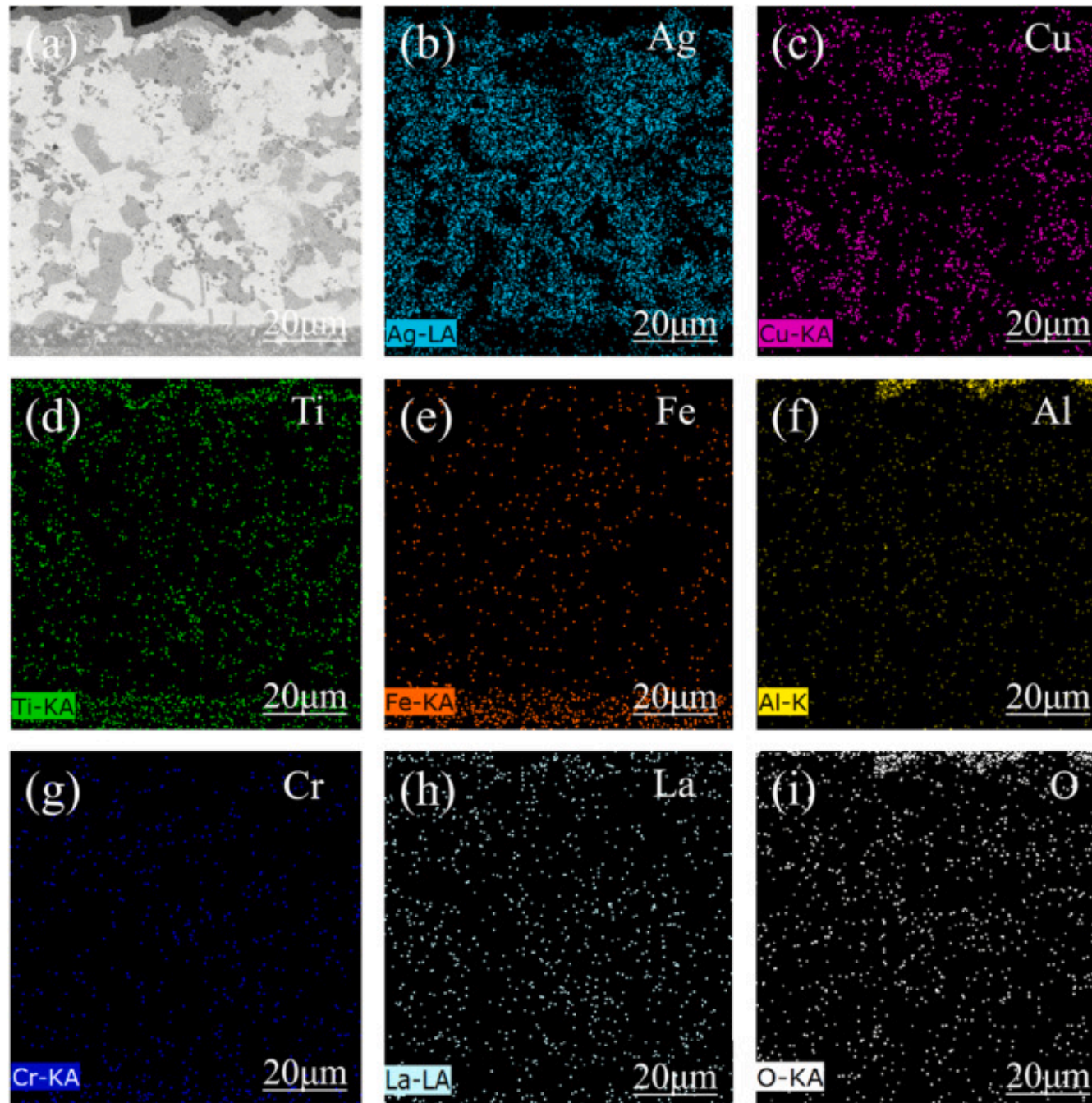
Based on a systematic analysis of the interfacial structure of Al₂O₃/316 SS joints brazed with AgCuTi single filler metal, an attempt was made to incorporate La into AgCuTi to improve the joint interfacial structure. Fig. 5 shows the typical microstructure of the Al₂O₃/316 SS joint brazed with AgCuTi-0.6La composite filler metal at 900°C for 10min. As shown in Fig. 5(a), the joint was well-bonded with no obvious defects. Similar to the joint obtained with AgCuTi single filler metal, the whole joint was divided into three regions: (I) the interfacial reaction layer on the Al₂O₃ side; (II) the Ag-Cu matrix and IMCs; and (III) the diffusion region on the 316 SS side. The chemical compositions of the various reaction phases in the joint were listed in Table 2. As shown in Fig. 5(b), a continuous and uniform Ti₃Cu₃O interfacial reaction layer (point 1) was still observed on the Al₂O₃ ceramic side in the joint brazed with AgCuTi-0.6La composite filler metal. The measured thickness of this reaction layer was approximately 2.0±0.2μm, which is slightly smaller than that in the joint brazed with AgCuTi single filler metal (2.3±0.2μm). This variation indicates that the introduction of La does not alter the dominant reaction pathway but rather modulates the growth kinetics of the existing reaction phases. More importantly, in region II, the originally coarse and aggregated Ti₂Cu and TiCu phases exhibited a finer and more dispersed distribution; in region III, the TiFe and TiFe₂ phases also tended to be refined.



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Fig. 5. Microstructure of joints brazed at 900°C for 10min: (a) Al₂O₃/AgCuTi-0.6La/316 SS joint; (b) Al₂O₃/AgCuTi-0.6La interface; (c) AgCuTi-0.6La/316 SS interface.



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Fig. 6. Elemental mappings of the Al₂O₃/AgCuTi-0.6La/316 SS joint brazed at 900°C for 10 min.

Table 2. EDS composition (at.%) results of the marked locations in Fig. 5.

Point	Ti	Cu	Fe	Ag	Al	Ni	Cr	La	Possible phases
1	50.96	29.20	-	0.21	8.39	-	-	1.07	Ti ₃ Cu ₃ O, Ti ₂ Cu
2	-	94.75	-	5.25	-	-	-	-	Cu(ss.)
3	-	3.82	-	96.18	-	-	-	-	Ag (ss.)
4	48.26	37.60	4.31	3.06	-	2.73	1.65	2.39	TiCu
5	36.06	0.60	52.54	0.09	-	3.10	5.83	1.78	TiFe ₂

The elemental mapping of the Al₂O₃/AgCuTi-0.6La/316 SS joint shown in Fig. 6 reveals that La was uniformly distributed within the joint and exhibited a high spatial correlation with Ti. Combined with the EDS analysis results of points 1, 4, 5 in Table 2 and the XRD analysis results in Fig. 9(e), it can be inferred that La did not form an independent new phase. Instead, it was incorporated into the Ti₃Cu₃O reaction layer and the Ti-Cu/Ti-Fe phase lattices in the form of solid solution, or segregated at phase boundaries. This feature reveals that the essence of La participation in the interfacial reaction is “modulatory” rather than “constructive”—that is, La does not replace Ti to form new phases, but rather regulates the interfacial reaction process at the atomic scale through doping, from the three dimensions of thermodynamics, kinetics, and electronic structure.

From a thermodynamic perspective, La possesses a stronger oxygen affinity than Ti. It preferentially combines with active oxygen at the interface, effectively reducing the local oxygen activity and oxygen vacancy concentration, thereby enhancing the densification and structural stability of the Ti₃Cu₃O reaction layer. Meanwhile, the differences in charge and ionic radius between La³⁺ (1.03 Å) and Ti⁴⁺ (0.61 Å) disrupt the charge balance of the original lattice [45], inducing local electron cloud rearrangement and enhanced bonding, which directly reduces the interfacial energy γ . According to classical nucleation theory, a decrease in interfacial energy γ significantly lowers the nucleation barrier: $\Delta G^* = \frac{16\pi\gamma^3}{3\sigma^2}$ (where γ is the specific surface energy, ΔG_v is the

Gibbs free energy difference per unit volume), thereby promoting the formation of more nuclei and achieving in-situ refinement and uniform precipitation of the reaction phases.

From a kinetic perspective, the “solute drag effect” of La is the key mechanism regulating the interfacial microstructure. The atomic radius of La (1.88 Å) is larger than that of Ti (1.32 Å) (as shown in Table 3). Its enrichment at the diffusion front creates an atomic-scale kinetic barrier [46], hindering the long-range Kirkendall diffusion of Ti and Cu atoms and retarding the growth rate of IMCs. Especially during the initial formation stage of Ti-Cu phases, the segregation of La at the growing interface disrupts the ordered atomic stacking, suppresses grain coarsening, and ultimately leads to the formation of fine and dispersed precipitates.

Table 3. Radius of each element in filler metal.

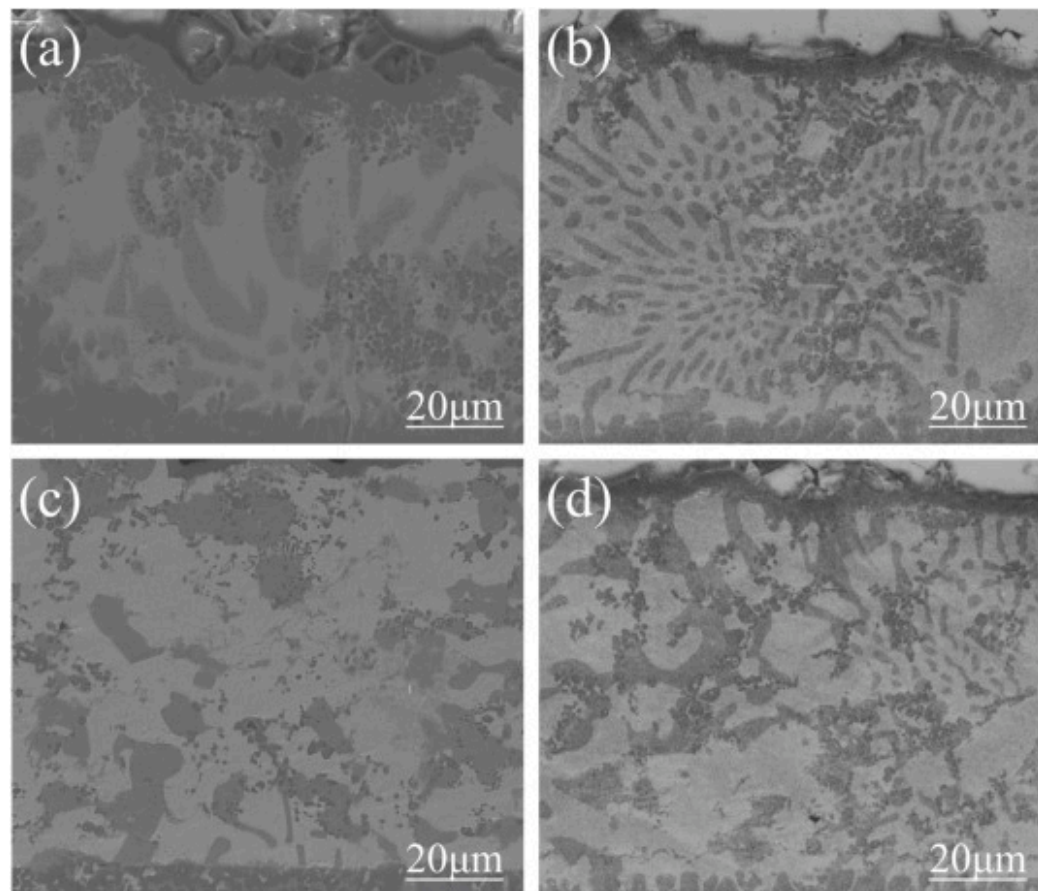
Element	Ti	Cu	Ag	La
Atomic radius/Å	1.32	1.17	1.34	1.88

Deeper electronic structure analysis reveals that the 5d/6s valence orbitals of La can effectively hybridize with the 3d orbitals of Ti, causing the characteristic peak of the Ti 3d density of states to shift toward the Fermi level and significantly increase in intensity. This change in electronic state enhances the overlap and hybridization between the Ti 3d orbital and the O 2p, Fe(Cu) 3d orbitals, thereby altering the covalency ratio of Ti-O and Ti-M (M=Fe, Cu) bonds, enhancing the directionality and strength of bonding, and optimizing the distribution and overlap of electron clouds in the interfacial region. Ultimately, this improves the interaction energy between interfacial atoms and strengthens the interfacial bonding.

3.4. Effect of La content on the microstructure of the Al₂O₃/316 SS brazed joints

Fig. 7 illustrates the effect of La content on the microstructure of the Al₂O₃/316 SS joints using AgCuTi-xLa filler metals at 900 °C for 10 min. It can be observed that as the La addition increased from 0 wt% to 0.6 wt%, the joint microstructure exhibited a distinct trend of refinement and homogenization, reflecting the critical role of La in regulating the interfacial reaction behavior. When the AgCuTi single filler metal was used (Fig. 7(a)), the joint primarily consisted of a continuously distributed Ag (ss.) and Cu(ss.) matrix. A relatively thick and non-uniform Ti₃Cu₃O interfacial reaction layer formed adjacent to the Al₂O₃ ceramic, exhibiting an irregular morphology with localized thickening, indicative of pronounced heterogeneous growth characteristics. The average

thickness of this layer was approximately $2.3 \pm 0.2 \mu\text{m}$. The formation of this reaction layer originated from the rapid diffusion of Ti atoms toward the ceramic side and their subsequent interfacial reaction with oxygen: $\text{L} + \text{Al}_2\text{O}_3 \leftrightarrow \text{Ti}_3\text{Cu}_3\text{O} + \text{Al}$. This process was driven by the local oxygen activity and the Ti chemical potential gradient, representing a typical in-situ interfacial reaction. In region II, substantial blocky or network-distributed Ti-Cu phases were observed. Concurrently, region III formed relatively large and densely distributed Ti-Fe phases. At a La content of 0.4wt% (Fig. 7(b)), the thickness of the $\text{Ti}_3\text{Cu}_3\text{O}$ interfacial reaction layer slightly decreases to $2.1 \pm 0.2 \mu\text{m}$. Its morphology became more uniform and continuous, indicating that La effectively suppressed the excessive enrichment of Ti and mitigated the intense interfacial reaction. Simultaneously, the proportion of the Ag (ss.) phase increased, and the Cu(ss.) phase was further refined. Additionally, the Ti-Fe phases in region III became locally discontinuous. These microstructural changes demonstrate the positive regulatory effect of La on the phase precipitation behavior. With an increase in La content to 0.6wt% (Fig. 7(c)), the interfacial reaction layer was further thinned to $2.0 \pm 0.2 \mu\text{m}$ while maintaining good continuity and compactness. This suggests that La not only inhibits undesirable interfacial side reactions but also promotes the formation of a more stable transition layer. At this stage, the Ti-Cu phases in region II were almost completely suppressed, appearing only sporadically as fine particles. Concurrently, the Ag-Cu phases formed an interwoven and interpenetrating gradient structure, while the discontinuity of the Ti-Fe compound layer in region III was further intensified, resulting in a loose, skeleton-like structure. However, when the La content reached 0.8wt%, the spreading ability of the composite filler metal deteriorated, and the thickness of the interfacial reaction layer rebounded to $2.2 \pm 0.2 \mu\text{m}$. Compared to the 0.6wt% La joint, the IMCs in region II exhibited a significant rebound, appearing as coarse, blocky aggregates. The skeleton-like structure of the Ti-Fe phases in region III gradually disappeared. This indicates that excessive La disrupts the kinetic balance of the interfacial reaction, leading to destabilized precipitation behavior.



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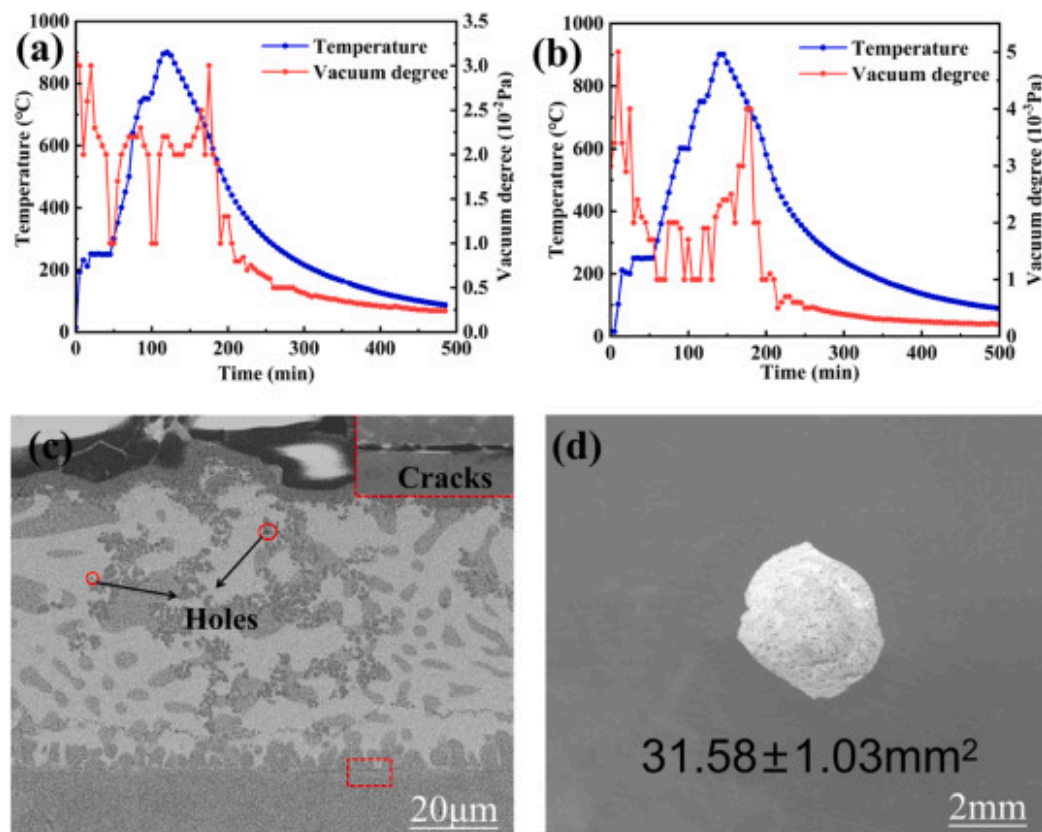
Fig. 7. Microstructures of Al₂O₃/AgCuTi-xLa/316 SS joints brazed at 900 °C for 10 min: (a) 0wt%, (b) 0.4wt%, (c) 0.6wt%, (d) 0.8wt%.

The partial molar enthalpies of the elements in the brazing seam are as follows: Ti-Cu: 10kJ/mol, Ti-O: 540kJ/mol, La-O: 1790kJ/mol [47,48]. These key thermodynamic data elucidate the essence of La's regulation of microstructural evolution: the precise modulation of the growth kinetics of IMCs based on the synergistic effect of thermodynamic priority and lattice compatibility. During the brazing process, La atoms competitively occupy oxygen-binding sites within the Ti₃Cu₃O lattice. However, constrained by the Hume-Rothery rule (the ionic radius of La³⁺ exceeds the 15% lattice tolerance limit relative to Ti⁴⁺) [49], La³⁺ cannot be stably incorporated into the Ti₃Cu₃O lattice to form a new phase. This incomplete substitution effect yields

two key consequences. First, modulation of the reaction driving force: the high oxygen affinity of La consumes active oxygen at the interface, reducing the local oxygen activity. This weakens the driving force for the reaction $\text{Ti} + \text{Cu} + \text{O} \rightarrow \text{Ti}_3\text{Cu}_3\text{O}$, which directly explains the observed decrease in reaction layer thickness with increasing La content in Fig. 7. Second, formation of a diffusion barrier: the embedding of La^{3+} into the $\text{Ti}_3\text{Cu}_3\text{O}$ lattice induces significant lattice distortion, generating elastic strain energy $\Delta G_{\text{strain}} = \frac{3}{2} V_{\text{m}} \epsilon^2 \mu$ (where V_{m} is the molar volume, ϵ is the strain, and μ is the shear modulus). This strain energy field creates an atomic-scale potential barrier that hinders the diffusion of Ti and Cu atoms. This is the fundamental reason for the evolution of Ti-Cu phases in region II from blocky to refined to dispersed particles. Nevertheless, when the La content reaches 0.8wt%, negative effects become dominant. Supersaturated adsorption of La at phase boundaries causes a nonlinear rebound in interfacial energy, counteracting the interfacial energy reduction effect achieved with an optimal La amount. Concurrently, elemental segregation induced by excessive La disrupts the uniformity of the diffusion field, leading to local coarsening and agglomeration of IMCs, thereby deteriorating the homogeneity of the microstructure.

3.5. Effect of vacuum degree on Al₂O₃/AgCuTi-0.6La/316 SS joint

Fluctuations in vacuum degree during brazing are normal, as moisture and organic compounds in the solder paste evaporate during the brazing process. Fig. 8(a) and (b) present the vacuum variation during brazing of the Al₂O₃/AgCuTi-0.6La/316 SS joint under 10^{-2} Pa and 10^{-3} Pa, respectively. The latter achieved a higher vacuum of 2×10^{-4} Pa, indicating a lower partial pressure of residual reactive gases in the furnace. This effectively suppressed oxidation of both the substrates and the filler metal and minimized interference from gaseous impurities in interfacial reactions. However, under 10^{-2} Pa, the higher residual gas facilitated bubble entrapment at the brazing interface, leading to porosity formation during cooling, as shown by the defects in Fig. 8(c).



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Fig. 8. Al₂O₃/AgCuTi-0.6La/316 SS joint brazed at 900°C for 10 min: (a) brazing process curve at 10⁻² Pa; (b) brazing process curve at 10⁻³ Pa; (c) microstructure of the joint brazed at 10⁻² Pa; and (d) spreading behavior of AgCuTi-0.6La filler metal on 316 SS at 10⁻² Pa.

The wetting morphology in Fig. 8(d) reveals that no wetting ring formed under low vacuum, indicating poor wettability. The diffusion area was only 31.58 mm², representing a 68.63% reduction compared with the higher vacuum. In this environment, residual gas impeded elemental diffusion between the base metal and the filler, preventing sufficient spreading of the filler metal. Concurrently, more Ti-Cu phases formed within the seam and concentrated in the seam center. The accumulation of these hard-brittle compounds increased localized residual stress, reduced elasticity in that region, and impeded the release of welding

thermal stresses, thereby weakening joint durability. Moreover, low-vacuum brazing caused significant cracking at the reaction interface between the filler metal and 316 SS, making cracks prone to propagation under external loading. In summary, the low-vacuum atmosphere was unfavorable for achieving reliable ceramic-metal brazed joints.

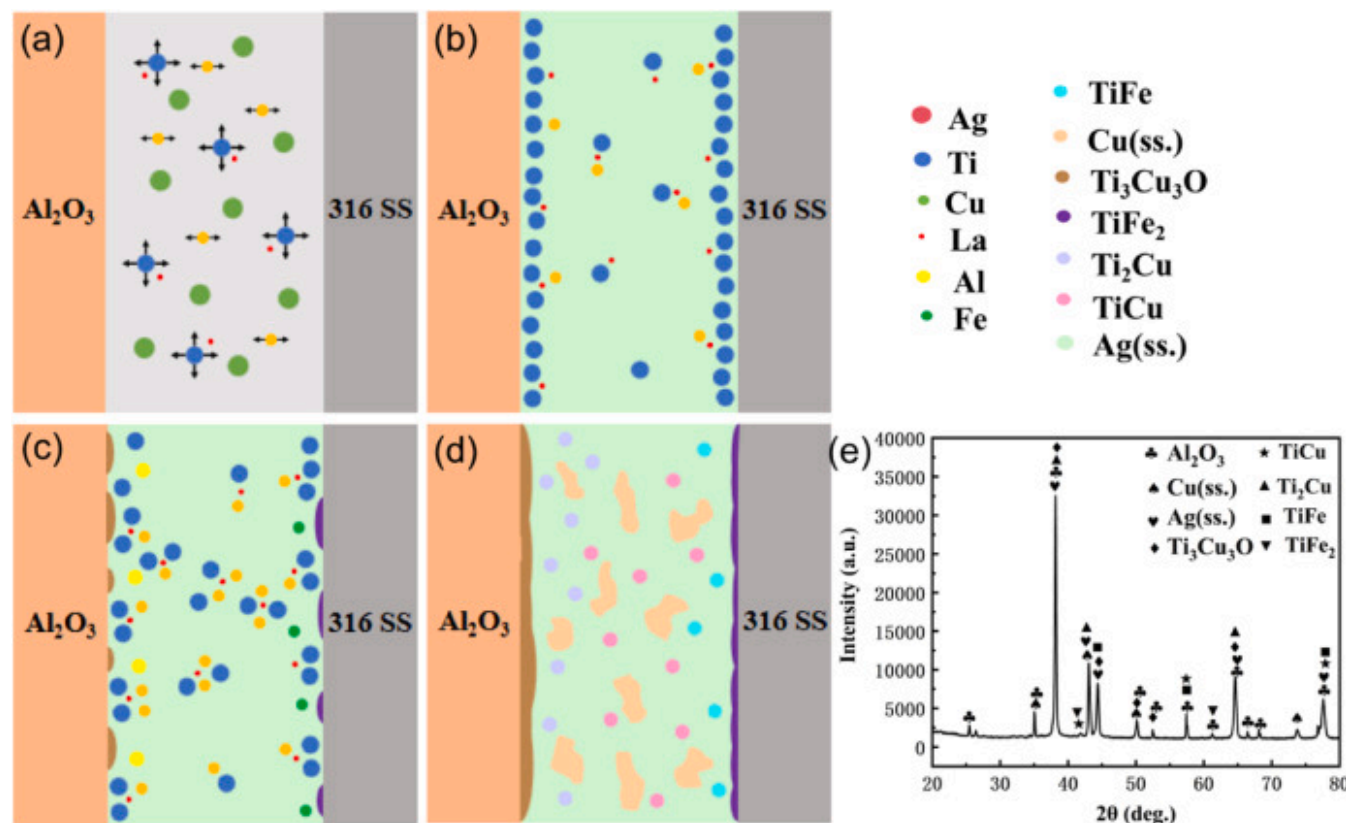
3.6. Interfacial evolution mechanism of the Al₂O₃/AgCuTi-0.6La/316 SS joint

Based on the aforementioned microstructural characterization and elemental diffusion analysis, [Fig. 9](#) proposes the interfacial evolution mechanism of the Al₂O₃/AgCuTi-0.6La/316 SS joint. The formation process of the entire joint can be divided into the following three stages:

- (1) Contact between the filler metal and substrates. As the brazing temperature increases, the AgCuTi-0.6La composite filler metal progressively softens and melts, establishing intimate contact with the substrates on both sides. During this stage, the filler metal remains in a solid or semi-solid state, and no significant elemental interdiffusion or chemical reaction has yet occurred. It primarily spreads across the interfacial region through physical wetting.
- (2) Interdiffusion and reaction of elements between the filler metal and the substrates. When the temperature rises above the eutectic point [[50](#)], the filler metal completely melts, entering a liquid flow stage. The active element Ti begins to rapidly diffuse toward both base materials. On the Al₂O₃ ceramic side, Ti preferentially adsorbs at defect sites on the ceramic surface, such as oxygen vacancies. Through charge transfer, it perturbs the stability of the Al-O lattice, promoting the rupture of some Al-O bonds. Based on thermodynamic priority, Ti combines with the released oxygen to form low-valence oxide TiO, which serves as an intermediate transition phase. This product subsequently undergoes an interfacial reaction with the adjacent Cu-rich region, forming a continuous and dense Ti₃Cu₃O composite oxide layer. The displaced Al atoms dissolve into the Ag (ss.) and Cu(ss.) matrices in the form of interstitial or substitutional solid solutions, without obvious segregation or precipitation of new phases. In the central region of the brazing seam, the short diffusion distance between Ti and Cu, coupled with high local elemental concentrations and a strong reaction driving force, thermodynamically favors the formation of Ti₂Cu and TiCu. The introduction of 0.6wt% La effectively hinders the long-range migration of Ti and Cu atoms, lowers the nucleation barrier, and promotes the generation of a large number of fine nuclei. Furthermore, owing to its chemical inertness (reflected in its high

reaction free energy with Ti and Cu) and its limited diffusion capability, Ag exists predominantly as a continuous matrix of the Ag (ss.) single phase within the brazing seam, thereby ensuring the toughness of the joint. On the stainless steel side, Fe undergoes only limited short-range diffusion, reacting with Ti near the interface to form the TiFe phase, which remains confined to the stainless steel surface without penetrating the center of the brazing seam. As the reaction proceeds, Ti further penetrates into the stainless steel substrate and reacts with Fe to form the thermodynamically more stable TiFe₂ phase, thereby establishing a “TiFe → TiFe₂” gradient transition diffusion zone.

- (3) Cooling and phase precipitation. During the cooling stage, the growth of the various reaction phases within the joint is essentially completed, and they precipitate sequentially according to their thermodynamic stability. The Ti₃Cu₃O reaction layer remains intact, serving as the critical bonding phase on the ceramic side. The Ti₂Cu and TiCu phases are further refined and become uniformly dispersed as the temperature decreases. The unreacted Ag-rich and Cu-rich phases precipitate, forming an interwoven Ag-Cu duplex structure that reinforces interfacial continuity. On the 316 SS side, the TiFe and TiFe₂ phases precipitate successively, forming a continuous skeleton-like structure. Ultimately, the interfacial structure of Al₂O₃/316 SS joints brazed at 900°C for 10min using AgCuTi-0.6La composite filler metal is as follows:
Al₂O₃/Ti₃Cu₃O/Ti₂Cu+TiCu/Ag (ss.)+Cu(ss.)/TiCu+TiFe/TiFe₂/316 SS.



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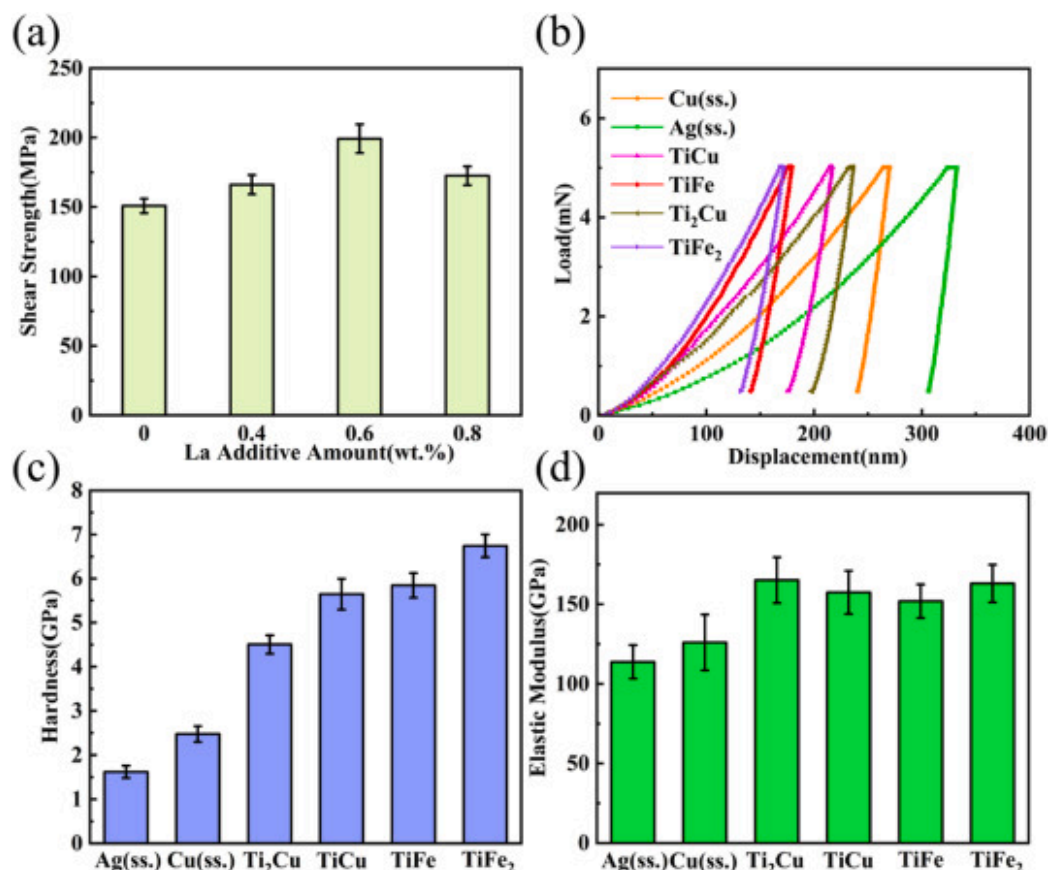
Fig. 9. Schematic illustration of the microstructural evolution in the Al₂O₃/AgCuTi-0.6La/316 SS joint brazed at 900°C for 10 min: (a) initial state, (b) melting of the filler metal, (c) elemental interdiffusion and reaction stage, (d) final microstructure after solidification; (e) XRD analysis of the Al₂O₃/AgCuTi-0.6La/316 SS joint.

3.7. Mechanical properties of the Al₂O₃/316 SS brazed joints

3.7.1. Shear strength of the brazed joints

Shear strength is a critical indicator for evaluating the quality of brazed joints, and its performance is co-determined by the interfacial reaction layers on both the ceramic and stainless steel sides [51], as well as the formation and distribution of IMCs

within the brazing seam [52,53]. Fig. 10(a) presents the effect of La content on the shear strength of Al₂O₃/316 SS joints using AgCuTi-xLa composite filler metals at 900°C/10min. Notably, the shear strength of the joints initially increases and then decreases as La content increases from 0wt% to 0.8wt%. The average shear strength of the AgCuTi joint without La addition is 150.85±5.25MPa. At 0.4wt% La, the strength increases to 166.15±6.95MPa, with a limited improvement in mechanical properties due to the insufficiently developed regulatory effect of La. The optimal La content is 0.6wt%, yielding an average shear strength of 199.25±10.35MPa, which represents a 32.08% increase compared with the AgCuTi joint. When the La content is further increased to 0.8wt%, the strength decreases to 172.55±6.75MPa, although it remains superior to that of the AgCuTi joint. Fig. 10(c) and (d) illustrate the hardness and elastic modulus distributions of the reaction phases within the Al₂O₃/316 SS joints, measured under a load of 5mN. The nano-hardness values are as follows: Ag (ss.) (1.621±0.14GPa), Cu(ss.) (2.476±0.18GPa), Ti₂Cu (4.506±0.21GPa), TiCu (5.644±0.35GPa), TiFe₂ (6.743±0.26GPa), and TiFe (5.843±0.28GPa). Notably, the phases containing Ti generally exhibit higher hardness.



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Fig. 10. Mechanical properties of the Al₂O₃/AgCuTi-xLa/316 SS brazed joints: (a) average shear strength of composite brazed joints with different La contents, (b) load-displacement curve, (c) nano-hardness, (d) elastic modulus.

The foregoing analysis indicates that the addition of an appropriate amount of La can lower the nucleation barrier, promote nuclei proliferation, and thereby achieve in-situ refinement and uniform dispersion of the reaction phases. On the one hand, these refined IMCs act as pinning agents, strengthening the solid solution matrix. The concomitant increase in grain boundary density impedes dislocation motion, directly enhancing the mechanical strength of the joint. On the other hand, the dispersed distribution of IMCs effectively mitigates local stress concentration, reducing the risk of crack initiation at stress raisers.

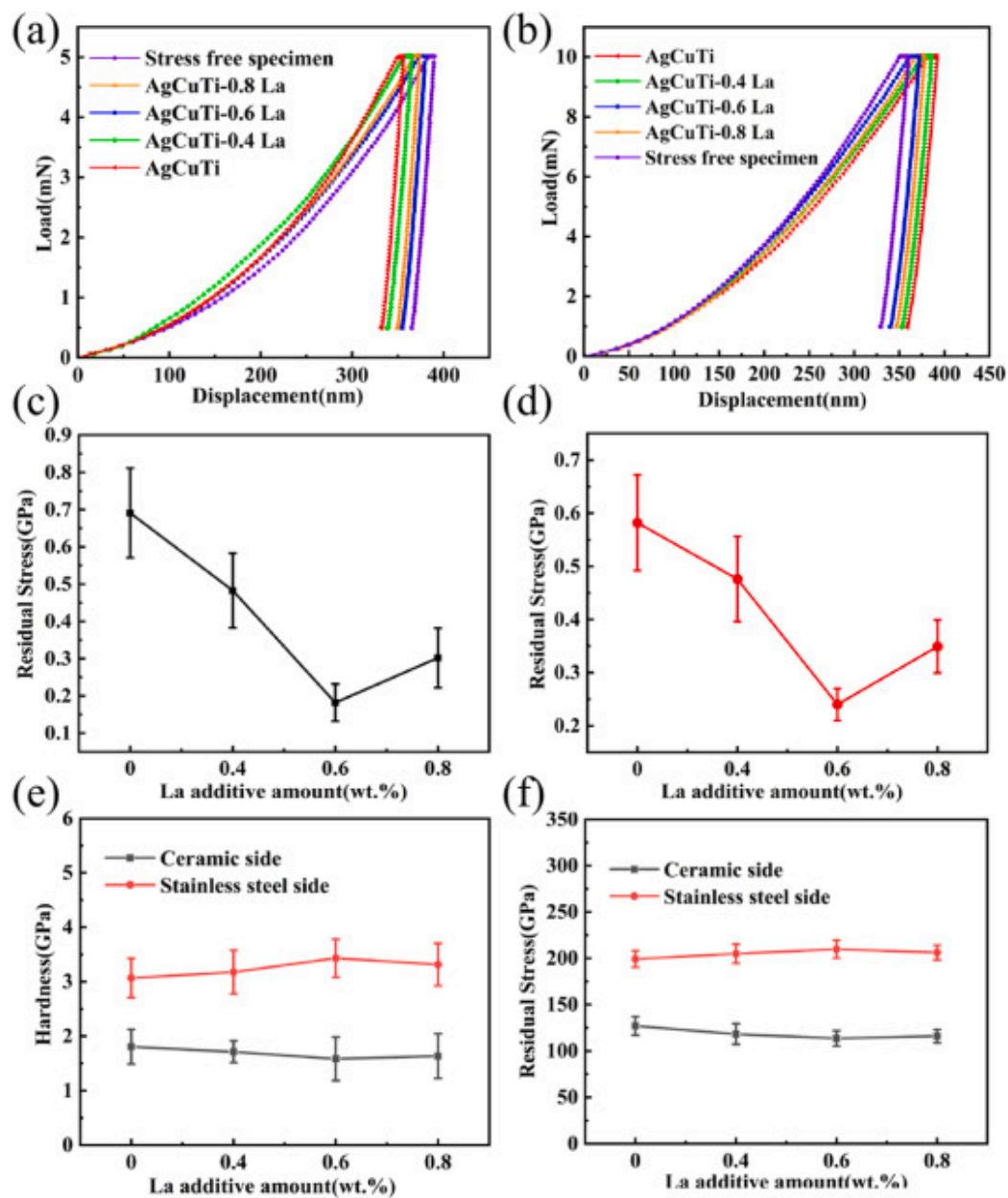
Furthermore, the segregation of La at phase boundaries, through the Gibbs adsorption effect, reduces interfacial tension. This alleviates the thermal expansion mismatch stress between Al₂O₃ ($8 \times 10^{-6}/^{\circ}\text{C}$) and 316 SS ($16 \times 10^{-6}/^{\circ}\text{C}$) [43,54], thereby improving the structural integrity of the joint. It should be noted that an excessively thick Ti₃Cu₃O reaction layer or excessive agglomeration of hard-brittle phases such as TiFe, TiFe₂, TiCu, and Ti₂Cu can restrict the plastic deformation capacity of Ag (ss.) and Cu(ss.). This hinders the relaxation of brazing residual stresses and exacerbates local stress concentration, ultimately predisposing the joint to cracking and premature failure [55,56]. The addition of La significantly suppresses the formation of such brittle IMCs, with this inhibitory effect being most pronounced at an optimal La concentration (approximately 0.6wt%).

3.7.2. Residual stress in the brazed joints

The significant mismatch in the coefficients of thermal expansion (CTE) between Al₂O₃ and 316 SS leads to incompatible shrinkage during the cooling stage of brazing. This thermally induced strain is constrained by the multi-layered structure of the joint and is further amplified by the elastic modulus mismatch among the ceramic, metal, and interlayer, ultimately resulting in a high residual stress field within the interfacial region. Based on previous studies [57] and comparative analyses of the CTEs among the ceramic, brazing filler metal, and metal [58], it can be inferred that the ceramic side of the joint primarily experiences residual compressive stress, whereas the stainless steel side is dominated by residual tensile stress. These internal stresses may induce microcrack initiation or accelerate its propagation under service loads, severely compromising the structural integrity of the joint.

Fig. 11(a) presents the load-displacement curves measured on the ceramic side of joints with different La contents, based on a stress-free AgCuTi/Al₂O₃ reference specimen. The indentation depth initially increases and then decreases with increasing La content. According to the residual compressive stress calculation formula (1) in Section 2.3, a larger indentation depth corresponds to a lower residual compressive stress. This phenomenon directly confirms the regulatory effect of La on the stress state on the ceramic side. Combined with the nanoindentation results of the ceramic side shown in Fig. 11(c)–(e), and (f), it was evident that in the absence of La, the nano-hardness on the ceramic side is $1.805 \pm 0.32 \text{ GPa}$, and the elastic modulus is $127.025 \pm 9.9 \text{ GPa}$. At this stage, the Ti-Cu hard-brittle phases were distributed as blocky aggregates within the Ag-Cu matrix, forming a distinct “hard-soft” abrupt interface that lacks an effective stress buffering zone. Simultaneously, the Ti₃Cu₃O reaction layer on the ceramic side exhibited significant brittleness, and its CTE is markedly mismatched with that of the ceramic substrate [59]. The excessively thick ($2.3 \pm 0.2 \mu\text{m}$) and irregularly shaped reaction layer exacerbated the stress accumulation caused by thermal mismatch, resulting in a high residual compressive stress of $0.691 \pm 0.12 \text{ GPa}$ on the ceramic side, which severely

deteriorated the mechanical properties of the joint. With the addition of 0.6wt% La, the Ti-Cu phases were refined into dispersed particles and uniformly distributed within the matrix. The grain boundary density increased significantly, forming an effective “stress buffering zone.” Meanwhile, the Ti₃Cu₃O reaction layer was uniformly thinned to $2.0 \pm 0.2 \mu\text{m}$, promoting smooth stress transfer across the ceramic/reaction layer/filler metal interfaces and avoiding local abrupt stress accumulation. These synergistic effects substantially alleviated the interfacial stress gradient, increased the tortuosity of crack propagation paths, and reduced the residual compressive stress to $0.182 \pm 0.05 \text{ GPa}$, a decrease of 73.66%. Correspondingly, the nano-hardness decreased to $1.583 \pm 0.4 \text{ GPa}$, and the elastic modulus decreased to $113.639 \pm 8.3 \text{ GPa}$, representing reductions of 12.30% and 10.54%, respectively, compared to the La-free group. According to the study by Yue et al. [60], relaxation of residual stress can promote dislocation slip and enhance the plastic deformation capacity of the material, thereby achieving a synergistic optimization of reduced residual stress and improved shear strength. However, when the La content was further increased to 0.8wt%, the nano-hardness on the ceramic side rebounded to $1.63 \pm 0.41 \text{ GPa}$, and the elastic modulus rebounded to $116.026 \pm 7.1 \text{ GPa}$. Supersaturated adsorption of excess La at phase boundaries inhibited the uniform diffusion of Ti and Cu atoms, leading to a re-emergence of compositional inhomogeneity within the brazing seam. Simultaneously, the thickness of the Ti₃Cu₃O reaction layer rebounded to $2.2 \pm 0.12 \mu\text{m}$, the original stress buffering effect disappeared, and the residual compressive stress increased again to $0.302 \pm 0.08 \text{ GPa}$, directly contributing to a decrease in the joint shear strength.



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Fig. 11. Nanomechanical properties of the Al₂O₃/AgCuTi-xLa/316 SS brazed joint: (a, b) load-displacement curves for the ceramic and stainless steel sides, (c, d) residual stress distribution on the ceramic and stainless steel sides, (e) nano-hardness, (f) elastic modulus.

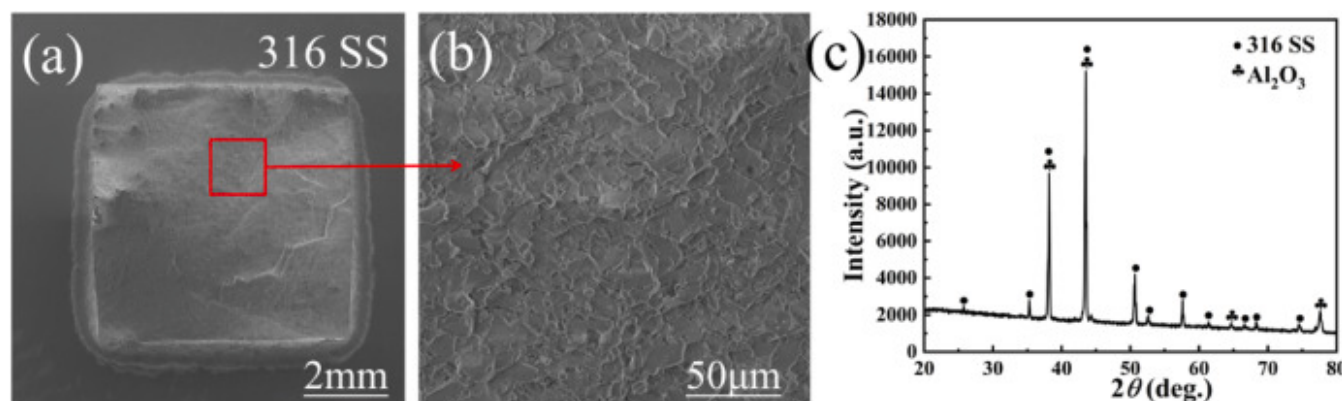
Fig. 11(b) shows the load-displacement curves measured on the stainless steel side of joints with different La contents, based on a stress-free AgCuTi/316 SS reference specimen. The results indicate that the indentation depth initially decreases and then increases with increasing La content, suggesting that the residual tensile stress on the stainless steel side first decreases and then rebounds. Combined with the nanoindentation results for the stainless steel side shown in Fig. 11(d), (e), and (f), in the absence of La, Ti segregated at the interfacial region, forming a densely distributed Ti-Fe hard-brittle phase lacking dispersion characteristics and a transitional structure. During cooling, stress concentrated at the phase interfaces, and coupled with the elastic modulus mismatch between the Ti-Fe phase and the matrix, this resulted in a high residual tensile stress of 0.582 ± 0.09 GPa on the stainless steel side, with a nano-hardness of 3.068 ± 0.36 GPa and an elastic modulus of 199.208 ± 0.89 GPa. With the addition of 0.6wt% La, Ti no longer segregated at the stainless steel side but instead diffused uniformly toward the center of the brazing seam, reacting with Cu to form fine, dispersed Ti-Cu phases. These replaced the original dense Ti-Fe phase, creating a loose, skeleton-type reaction zone conducive to stress relief. The increased grain boundary density also facilitated cooling deformation through grain boundary sliding. Furthermore, the pre-brazing solution treatment effectively suppressed the outward diffusion of Fe, further reducing the content of the Ti-Fe phase. Although the nano-hardness increased to 3.431 ± 0.35 GPa (an 11.83% increase), the elastic modulus only slightly rose to 209.792 ± 0.96 GPa, without a significant abrupt change, indicating improved modulus matching between the hard and brittle phases and the matrix. Ultimately, the residual tensile stress on the stainless steel side decreased to 0.24 ± 0.03 GPa, a reduction of 58.76%. This significant stress relief, combined with the enhanced hardness from dispersion strengthening, created a synergistic effect that laid the foundation for the overall improvement in joint strength. When the La content was increased to 0.8wt%, the excess La induced re-agglomeration of the Ti-Fe phases, disrupting the stable skeleton structure formed at 0.6wt% La. The stress buffering space disappeared, and the aggregation of hard-brittle phases led to uneven stress transfer. Correspondingly, the hardness on the stainless steel side decreased to 3.315 ± 0.39 GPa, the elastic modulus dropped to 206.228 ± 8.1 GPa, and the residual tensile stress rebounded to 0.349 ± 0.05 GPa, further exacerbating the degradation of joint performance.

In summary, the introduction of La synergistically regulates the interfacial residual stress state through multiple mechanisms. First, the refinement and dispersed distribution of phase morphology disrupt the continuous network of the original brittle phases, creating numerous grain boundaries and phase boundaries that serve as stress relaxation channels. Second, the

homogenization and controlled critical thickness of the interfacial reaction layer effectively alleviate stress concentration caused by CTE and elastic modulus mismatches, enabling progressive load transfer. Finally, the smoothing of the elastic modulus gradient reduces abrupt stiffness changes at the interface, promoting a more homogeneous stress field. However, hindered diffusion and phase separation induced by excessive La disrupt this balance, leading to a decline in performance.

3.7.3. Fracture mode of the brazed joints

The fracture surface morphology of the joints was characterized by SEM, and the fracture path and failure location were determined by combining XRD and EDS analyses. The fracture morphology of the joint brazed with the AgCuTi single filler metal is shown in Fig. 12. The joint fracture occurred entirely within the Al₂O₃ ceramic substrate, and the fracture surface exhibited a typical step-like morphology. Fracture initiated in the ceramic substrate, where residual tensile stress concentration arose due to the CTE mismatch between the ceramic and stainless steel, compounded by the modulus difference between the Ti₃Cu₃O reaction layer and the ceramic matrix. Under external loading, stress drove crack propagation within the ceramic along a path perpendicular to the loading direction, ultimately leading to brittle fracture of the ceramic matrix. XRD analysis in Fig. 12(c) detected only diffraction peaks corresponding to Al₂O₃ and the 316 SS substrate, further confirming that fracture occurred within the ceramic matrix.

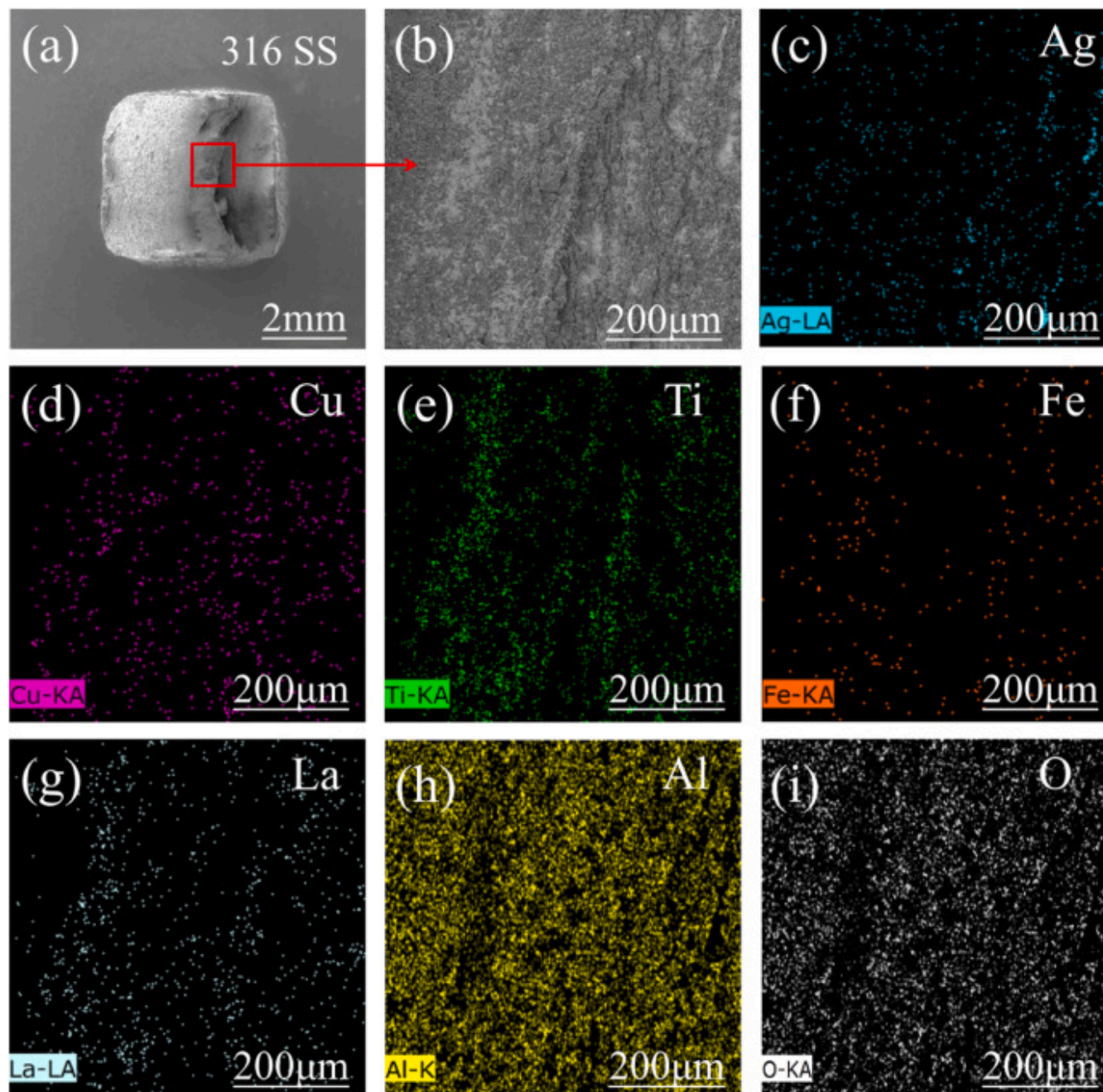


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Fig. 12. Al₂O₃/AgCuTi/316 SS joint: (a-b) fracture surface morphology, (c) XRD pattern of the fracture surface.

The fracture morphology of the joint brazed with the AgCuTi-0.4La composite filler metal is shown in [Fig. 13](#). The addition of La initially modified the seam microstructure, leading to a certain degree of refinement of the Ti-Cu phases, although a uniformly dispersed distribution was not achieved. The Ti₃Cu₃O reaction layer on the ceramic side remained brittle, and its thickness was not effectively controlled. Although the residual stress at the interface was slightly reduced, it remained concentrated in the reaction layer region. Consequently, under external loading, the relatively brittle Ti₃Cu₃O reaction layer became the preferred site for crack initiation. The residual tensile stress further drove crack propagation within the reaction layer, ultimately resulting in joint failure along the reaction layer. XRD analysis in [Fig. 16\(e\)](#) detected diffraction signals corresponding to Ti₃Cu₃O, Ag (ss.), and Cu(ss.) phases. Elemental mapping revealed a layered distribution of Ti and Cu, and only trace amounts of Ag were detected, confirming that the fracture path was primarily confined to the reaction layer on the ceramic side and the adjacent brazing seam region.

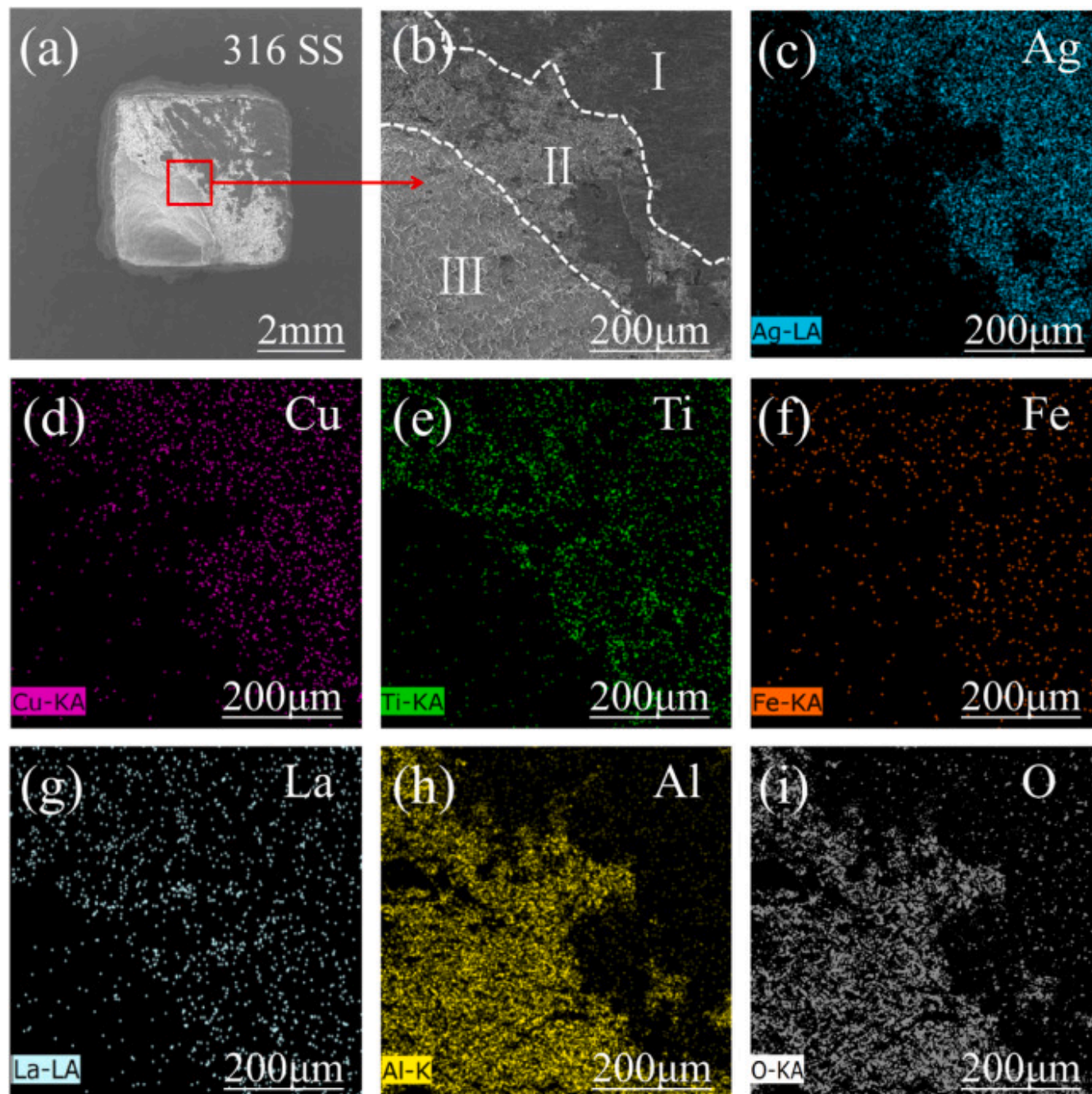


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Fig. 13. Al₂O₃/AgCuTi-0.4La/316 SS joint: (a-b) fracture surface morphology, (c-i) EDS elemental distribution on the fracture surface.

The fracture morphology of the joint brazed with the AgCuTi-0.6La composite filler metal is shown in Fig. 14. It can be observed that most of the brazing seam region was present on the fracture surface, with only a small amount of the Al₂O₃ substrate retained. The addition of La achieved significant refinement and redistribution of the weld microstructure: the agglomerated Ti-Cu phases on the ceramic side were refined into dispersed particles, forming a stress buffering zone; the thickness of the Ti₃Cu₃O reaction layer was uniformly reduced; the hard-brittle Ti-Fe phases on the stainless steel side were replaced by dispersed Ti-Cu phases. Consequently, the overall residual stress of the joint was substantially reduced and uniformly distributed, while the interfacial bonding strength between the reaction layer and the ceramic was simultaneously enhanced. Combined with the XRD analysis in Fig. 16(f), under this regulated stress and phase distribution, the crack initiation site shifted from the highly stressed interior of the ceramic to the Ag (ss.) region within the brazing seam. The crack propagated tortuously through the brazing seam, the reaction layer on the ceramic side, and the alumina ceramic substrate. The fracture surface could be divided into the brazing seam region (Region I), the reaction layer region on the ceramic side (Region II), and the Al₂O₃ ceramic region (Region III). The brazing seam and reaction layer regions exhibited quasi-cleavage fracture characteristics, while the ceramic region displayed typical brittle fracture features.

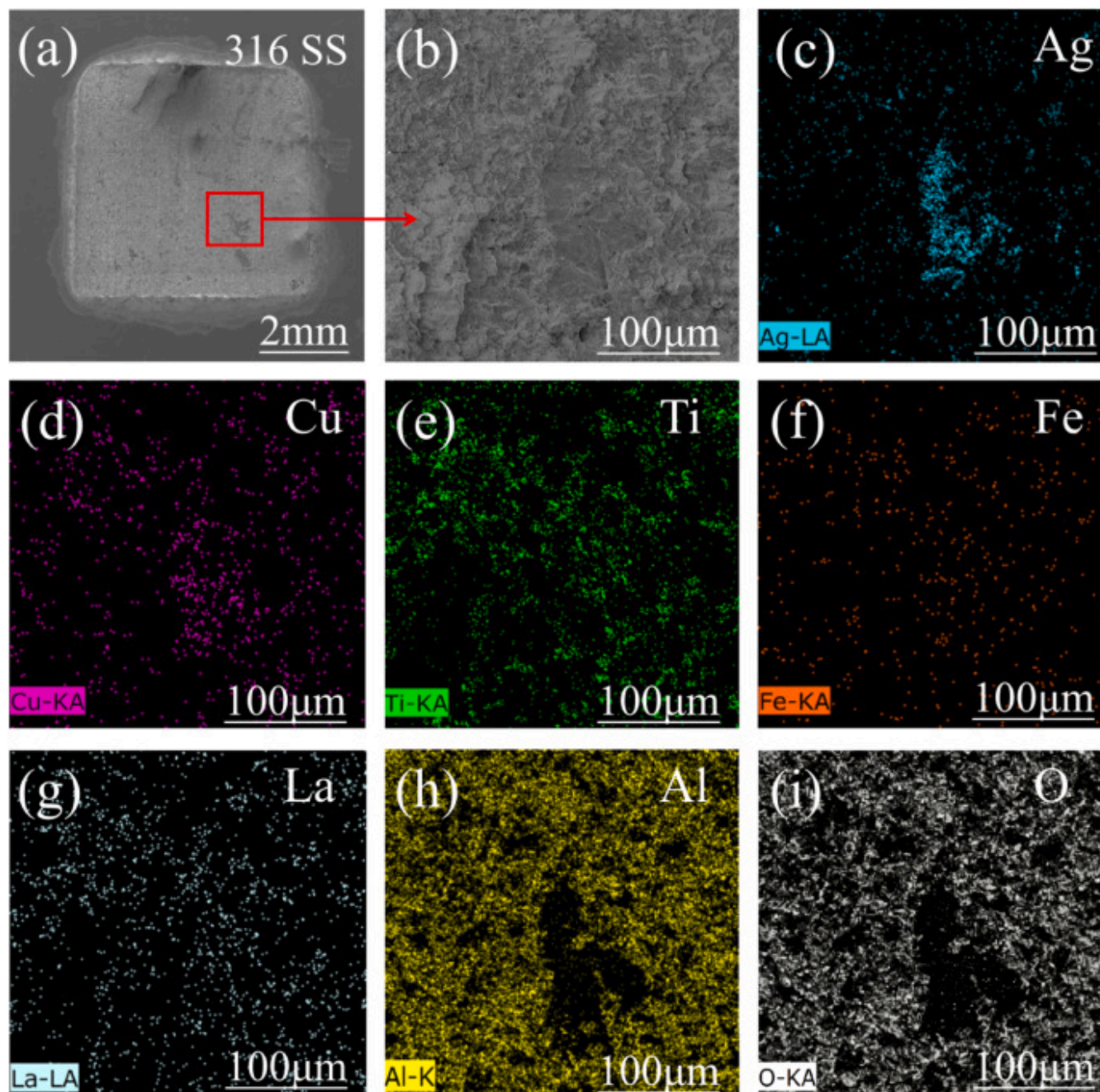


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Fig. 14. Al₂O₃/AgCuTi-0.6La/316 SS joint: (a-b) fracture surface morphology, (c-i) EDS elemental distribution on the fracture surface.

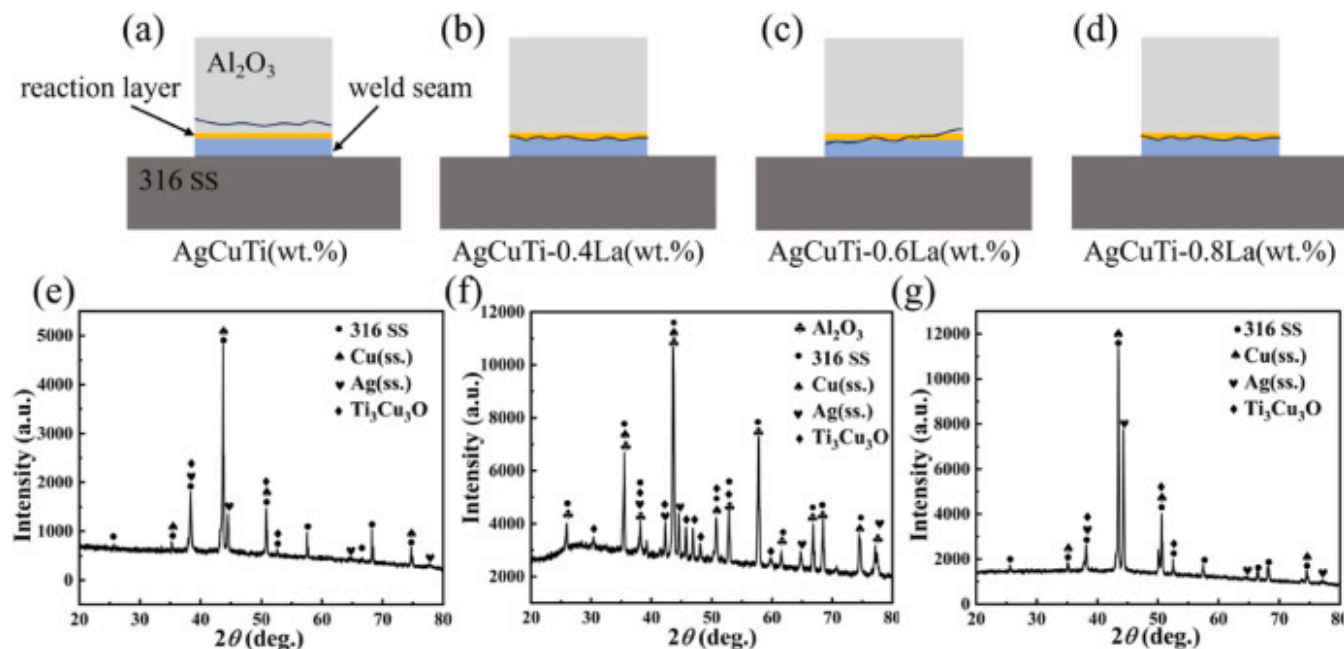
Fig. 15 shows the fracture morphology of the joint brazed with the AgCuTi-0.8La composite filler metal. Its fracture path was similar to that of the joint with 0.4wt% La. An excessively high La content significantly reduced the fluidity of the molten filler metal. The thickness of the Ti₃Cu₃O reaction layer rebounded, and the interfacial residual stress increased again with uneven distribution. Combined fracture surface XRD analysis (Fig. 16(g)) and elemental mapping revealed that although the fracture surface was still dominated by the reaction layer, Ag (ss.) was clearly present in the microstructure, indicating a more complex and diverse crack propagation path in the 0.8wt% La joint.



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Fig. 15. Al₂O₃/AgCuTi-0.8La/316 SS joint: (a-b) fracture surface morphology, (c-i) EDS elemental distribution on the fracture surface.



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Fig. 16. Fracture modes of the brazed joints: (a) AgCuTi, (b) AgCuTi-0.4La, (c) AgCuTi-0.6La, (d) AgCuTi-0.8La; and the corresponding XRD patterns of the fracture surfaces for joints brazed with (e) AgCuTi-0.4La, (f) AgCuTi-0.6La, and (g) AgCuTi-0.8La.

The essence of La's regulation of the fracture behavior of Al₂O₃/316 SS joints lies in the synergistic thermodynamic-kinetic modulation at the atomic scale. In the absence of La, uncontrolled diffusion led to the agglomeration of IMCs; the Ti₃Cu₃O reaction layer was thick with high interfacial energy; compounded by CTE mismatch, stress concentrated within the ceramic matrix, inducing brittle fracture within the ceramic. At 0.4wt% La, the solute drag effect initially suppressed long-range atomic diffusion, but phase refinement was uneven, transferring stress concentration to the brittle reaction layer, resulting in reaction layer-dominated fracture. At 0.6wt% La, by preferentially combining with active oxygen at the interface, La reduced local oxygen

activity, modulated the lattice electron cloud distribution, and simultaneously suppressed the coarsening of IMCs, forming a stress buffering zone. This significantly relaxed residual stress, leading to tortuous crack propagation across multiple regions and a mixed ductile-brittle fracture mode, with a 32.08% increase in shear strength. At 0.8wt% La, excess La induced a nonlinear rebound in interfacial energy and a diffusion barrier effect, causing phase re-agglomeration and stress rebound, reverting fracture to a reaction layer-dominated mode. The core mechanism is that La optimizes interfacial mechanical behavior by modulating phase evolution and stress distribution. Therefore, controlling the in-situ reactions of Ti with Cu and Fe to regulate the microstructural distribution of the joint is crucial for achieving robust Al₂O₃/316 SS joints.

3.8. Finite element analysis of residual stress in the Al₂O₃/AgCuTi-xLa/316 SS joints

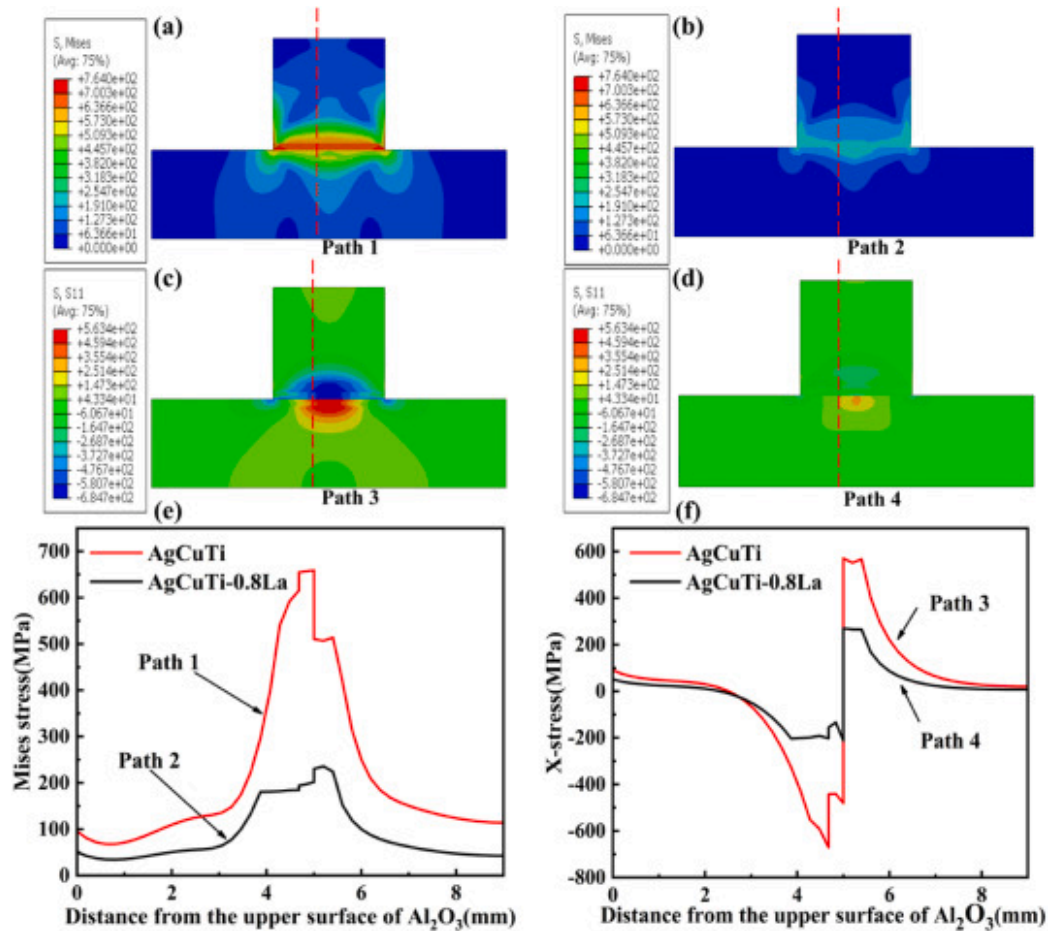
Most existing finite element simulations of residual stress in brazed joints primarily focus on global joint models, offering limited insight into internal stress states. Given this, this study concentrated on analyzing the stress distribution within the joint's central cross-section. The experimental results from the preceding sections indicated that La addition promotes inter-element diffusion, significantly reduces the content of the hard-brittle Ti-Cu phases, and facilitates their uniform distribution, thereby mitigating stress concentration. To substantiate this mechanism, a two-dimensional static simulation of the residual stress distribution in the joint was conducted using COMSOL software. Given the relatively small weld thickness in practical applications, which complicates accurate stress analysis, the weld thickness was appropriately increased in the model to facilitate subsequent finite element simulations. The simulation was simplified to a thermal-stress analysis, in which a thermal load was applied, and the assembly was uniformly cooled from the solidus temperature of the filler metal (780°C, set as the stress-free temperature) to 20°C, simulating the post-brazing cooling process. Fixed constraints were applied at the model boundaries to prevent rigid body displacement, while other surfaces remained unconstrained to accommodate thermal expansion. A graded mesh was implemented, with element sizes of 0.2mm for the ceramic and brazing layer and 0.5mm for the stainless steel. Common nodes at the interfaces ensured ideal bonding, thus achieving an optimal balance between computational accuracy and efficiency. Given that the actual reaction layer is extremely thin, the model adopted an “equivalent property layer” strategy, wherein the mechanical effects of the interfacial reaction layer and hard-brittle phases were quantified by adjusting the equivalent elastic modulus and yield strength of the brazing layer. The basis for material parameter selection is as follows: The CTE values of Al₂O₃ and 316 SS are $8 \times 10^{-6}/^{\circ}\text{C}$ and $16 \times 10^{-6}/^{\circ}\text{C}$, respectively. Y.H. Zhou [61] obtained the CTE of the TiCu phase using the following equation: $\alpha = \alpha_{\text{Ti}} V_{\text{f}} + (1 - V_{\text{f}}) \alpha_{\text{Cu}}$ (where α is the CTE of the TiCu phase in the brazing seam, and V_{f} is the volume fraction of Ti in the TiCu phase), the calculated CTE values of the TiCu, Ti₂Cu, TiFe, and TiFe₂ phases are approximately $12.1 \times 10^{-6}/^{\circ}\text{C}$, $11.5 \times 10^{-6}/^{\circ}\text{C}$, $10.9 \times 10^{-6}/^{\circ}\text{C}$, and $11.3 \times 10^{-6}/^{\circ}\text{C}$, respectively. The joint was modeled as a three-layer

structure comprising a 5 mm wide×5 mm thick ceramic layer, a 5 mm wide×0.3 mm thick weld layer, and a 16 mm wide×4 mm thick stainless steel layer. Based on this, the effect of La on the residual stress of the joint was systematically evaluated by comparing and analyzing the stress distribution in the AgCuTi and AgCuTi-0.6La joints.

The rationality of the simulation model is based primarily on the following mechanisms: the coarsening and segregation of hard-brittle phases such as TiCu raise the local elastic modulus in the weld region, as evidenced by nanoindentation results (~150 GPa). La addition counteracts this effect by refining the microstructure, reducing the hard-phase content, and improving its spatial uniformity, thereby lowering the joint's equivalent elastic modulus. According to grain refinement strengthening theory, such a refined and homogeneous microstructure not only strengthens the weld but also minimizes the property mismatch with the base metal, thus alleviating stress concentration. The accompanying increase in grain boundary density enhances dislocation pinning and crack path tortuosity, which further suppresses crack propagation. Moreover, grain boundary slip facilitates coordinated plastic deformation, contributing to stress homogenization. Finally, based on composite CTE principles [62], the uniform distribution of fine hard-brittle phases reduces the effective thermal expansion mismatch between the weld and base materials, thereby diminishing residual stress.

The residual stress distribution at the joint center cross-section is shown in Fig. 17. Compressive stresses were concentrated on the Al₂O₃ side, whereas tensile stresses were localized on the 316 SS side. This result validates the speculation regarding the stress states of the two substrates in Section 3.7.2. A comparison between Fig. 17(a) and (b) reveals that the addition of La significantly alleviated the overall residual stresses in the joint. According to Fig. 17(e), stress concentrations were predominantly located at the interface between dissimilar materials. The addition of La shifted the maximum stress location from the ceramic side to the stainless steel side, which explains the transition in the joint fracture mode from fracture within the ceramic (Type I) to mixed fracture (Type III). On the ceramic side, La mitigated the high stress concentrations caused by the CTE mismatch by decreasing the content of hard-brittle Ti-Cu phases, promoting their uniform distribution toward the seam center, and thinning the Ti₃Cu₃O reaction layer. On the stainless steel side, despite a smaller CTE mismatch with the filler metal, this region became a new relative stress concentration zone due to alterations in overall weld strength and more complete stress relaxation on the ceramic side. Fig. 17(c) shows the stress distribution along path 3 within the central cross-section of the Al₂O₃/AgCuTi/316 SS joint. Combined with Fig. 17(f), the residual compressive stress gradually increased from the top to the bottom of the ceramic, reaching a maximum of 684.301 MPa. Owing to its superior capacity for plastic deformation, the weld zone accommodated a portion of the stresses, leading to a reduction in the stress magnitude in its vicinity. Subsequently, alternating tensile and compressive stresses occurred at the weld center. The peak tensile stress on the stainless steel weld side reached 563.388 MPa,

and gradually decreased with increasing distance from the weld. Because the property mismatch between the filler metal and stainless steel was less severe than that with the ceramic, no distinct dual tensile stress center developed. Fig. 17(d) shows the stress distribution along path 4 in the central cross-section of the Al₂O₃/AgCuTi-0.6La/316 SS joint. Comparing Fig. 17(c), (d) and (f) reveals that the tensile-compressive stress patterns on both sides remained consistent after La addition. However, the residual compressive stress at the ceramic/weld interface decreased to 212.078MPa, representing a 69.01% reduction compared to the single filler metal joint. Similarly, the residual tensile stress at the weld/stainless steel interface diminished to 269.411MPa, corresponding to a 52.18% reduction compared to the single filler metal joint.



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Fig. 17. Stress distribution in finite element simulations: (a, b) Mises stress distribution for Al₂O₃/AgCuTi/316 SS and Al₂O₃/AgCuTi-0.6La/316 SS joints; (c, d) Stress distribution in the XX direction for Al₂O₃/AgCuTi/316 SS and Al₂O₃/AgCuTi-0.6La/316 SS joints; (e, f) Stress distribution along Paths 1, 2 and Paths 3, 4.

To validate the credibility of the model, this study compared the simulation results with experimental data. The simulation showed a significant reduction in interfacial residual stress after the addition of La (compressive stress decreased by 69.01%, tensile stress by 52.18%), which aligns well with the trends measured by nanoindentation (compressive stress decreased by 73.66%, tensile stress by 58.76%). This consistency confirms that La effectively mitigates local stress concentration and demonstrates the reliability of the model results.

4. Conclusion

This study employed vacuum brazing to join Al₂O₃ and 316 stainless steel using AgCuTi filler metals with varying La additions (0-0.8wt%). The influence of La on the interfacial microstructure and mechanical properties of the joint was systematically investigated. The main findings are summarized as follows:

- (1) The addition of La improved the wettability of the filler metal and optimized the joint microstructure. With 0.6wt% La, the spreading area increased by 31.80% to $100.68 \pm 1.03 \text{ mm}^2$, establishing a typical interfacial microstructure of Al₂O₃/Ti₃Cu₃O/Ti₂Cu+TiCu/Ag (ss.)+Cu(ss.)/TiCu+TiFe/TiFe₂/316 SS.
- (2) La enhanced the shear strength and promoted a more favorable stress distribution across the joint. At 0.6wt% La content, the joint shear strength increased by 32.08%, concurrently reducing residual compressive stresses at the ceramic interface by 73.66% and tensile stresses at the steel side by 58.76%.
- (3) La addition altered the fracture path from a brittle mode within the Al₂O₃ to a mixed mode that initiated in Ag (ss.), propagated through the Ti₃Cu₃O layer, and terminated in the Al₂O₃ matrix.
- (4) La promoted the diffusion and uniform redistribution of Ti and Cu, thereby suppressing Ti segregation and reducing the formation of hard-brittle Ti-Cu phases. This effect refined the interfacial microstructure, thereby alleviating the residual stresses induced by CTE mismatch and significantly enhanced the joint shear strength.

CRediT authorship contribution statement

Xinyu Zhou: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Zheye Liu:** Writing – review & editing, Investigation, Data curation. **Lingling Yang:** Writing – review & editing. **Yifei Guo:** Investigation. **Xin Liu:** Methodology, Formal analysis. **Chengshuang Zhou:** Writing – review & editing. **Lin Zhang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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